Kenya Agroecology Hub

Manor House Agricultural Centre





Field Assessment Report

Improving Farmer Training on AE Practices Through Collaboration and Follow-up

Project funded by
CONSERVATION FOOD AND HEALTH FOUNDATION

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Acronyms

AE	-	Agroecology
CFHF	-	Conservation,Food and Health Foundation
CREPP	-	Community Rehabilitation and Environmental Protection Program
DIG	-	Development In Gardening
FIPS	-	Farm Inputs Promotions
GIZ	-	Deutsche Gesellschaft fuer Internationale Zusammenarbeit (German development agency)
KALRO	-	Kenya Agriculture and livestock Research organization
KCWCG	-	Kipsaina Crane and wetland Conservation Group
MHAC	-	Manor house Agricultural Center
PI	-	Principal Investigator (Co-PI: Co-Principal Investigator)
RFRN	-	Rural Farmers Researrch Network
RODI	-	Resource Orientd Development Initiative
SOFDI	-	Sustainable Organic Farming and Development Innitiative
TAPE	-	Tool for Agroecology Performance Evaluation
TEMBEA	-	Tembea Youth

Acknowledgements

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In particular, the willingness of the farmers in the 42 sub counties to meet the endline evaluation team and participate in the survey and in-depth interviews to share their progress on adoption of AE practices was very important for this report. We would also like to acknowledge the financial support of the Conservation Food and Health Foundation. Without the commitment of the AE hub Co-PI's,Dr. Beth Medvecky and Mr. David Mwangi, who facilitated necessary logistics and technical support to review the tools and methods used in this evaluation, this work would not have been possible.

Executive Summary

Background and overview

The Kenya Agroecology Hub, anchored at Manor House Agricultural Centre (MHAC), implemented the project "*Improving Farmer Training on AE Practices Through Collaboration and Follow-up*" with support from the Conservation, Food and Health Foundation. The project brought together ten partner organisations working across ten counties and 42 sub-counties in western Kenya. It sought to strengthen the impact of one-week agroecology (AE)[1] training by assessing which practices farmers continued to apply after training and identifying factors influencing their adoption.

Purpose of the programe

The programme was designed to strengthen the practical application of AE knowledge beyond the training workshop. Farmers received training and were then given approximately one and a half seasons to apply the practices on their farms. The follow-up assessment provided an opportunity for MHAC and partner organisations to jointly examine what farmers were implementing and use the findings to improve future trainings and follow-ups.

Programme Beneficiaries

The programme involved 223 farmers across five training cohorts, with farmers mobilised through the ten partner organisations. Farmers were expected to continue learning and sharing experiences through groups after the training. The assessment also highlighted differences in farmers' ability to adopt practices, particularly where access to land, inputs, labour, assets, weather conditions and household decision-making affected implementation.

Programme Activities and Practices Taught

Training covered five thematic areas: soil health, pest management, nutrition and value addition, livestock production, and climate change adaptation. Across these areas, 63 practices were assessed during the baseline and endline[2] assessments. The practices included composting, botanical pest management, crop rotation, post-harvest handling, livestock management, water conservation and other AE approaches.

Purpose of the Evaluation

The evaluation assessed changes in farmers' reported and observed application of the practices taught during the training. It also sought to understand why some practices were adopted, maintained or discontinued, and whether the practices and training approaches were sufficiently relevant and achievable within farmers' circumstances.

Methodology

Data were collected through baseline and endline assessments, supported by field observation and follow-up discussions with farmers. The endline assessment was conducted jointly by MHAC and partner organisation staff. The analysis compared practice-level changes between the two assessment points and considered farmers' explanations and field observations alongside the quantitative results.

1 **Agroecology (AE).** Farming that applies ecological principles and works with local resources, biodiversity and farmers' own knowledge rather than relying mainly on bought inputs.
2 **Baseline and endline.** The baseline assessment recorded the practices farmers already used during the workshop. The endline assessment followed after farmers had about one and a half seasons to apply the practices.

Evaluation Analysis

The analysis examined changes across the 63 practices, including practices that were continued, newly adopted, discontinued or not adopted. Quantitative findings were interpreted alongside qualitative information and observations from farm visits. This was important because the assessment measured whether a practice was present, rather than the extent or quality of its implementation. The analysis also considered other factors that may have influenced adoption, including support received from partner organisations and other programmes.

Evaluation Findings

Overall, farmers reported increased application of most of the practices assessed. Forty-eight of the 63 practices increased between baseline and endline, with the strongest gains generally observed in practices requiring relatively few external resources, such as composting, botanical brews, crop rotation and post-harvest handling. However, adoption was constrained by access to seed and other inputs, land and assets, competing household needs, social factors and weather-related risks.

The findings also point to a gap between some training assumptions and conditions on farmers' farms. Practices such as vermicomposting[3], azolla[4] production and the introduction of new crops were not always easy for farmers to implement without additional materials, equipment or follow-up support. In addition, some reported changes could not be attributed solely to the training because farmers had received support from other programmes or had already been practising some of the assessed practices.

Recommendations

The findings suggest that improving adoption will require more than increasing the amount of information delivered during training. Future training should focus on fewer practices, with more time for practical demonstration and discussion of affordable alternatives. The programme should also strengthen access to relevant inputs and materials, pilot practices that require additional resources before wider promotion, and strengthen systematic follow-up by partner organisations during the first season after training. Future assessments should improve the measurement of the extent of practice adoption and better capture other support received by farmers.

Overall, the assessment supports the continued use of the joint MHAC-partner follow-up approach, while pointing to the need for stronger practical support between training and adoption.

3 Vermicompost. Compost made with the help of earthworms.
4 Azolla. A small floating water fern grown in ponds and used as livestock feed.



SECTION 1

Introduction (Context, Purpose and Methodology)

Photo 1. *Food forest farm tour: MHAC Director Mr David Mwangi sharing with AE Hub Community of Practice (CoP) partners during the joint learning workshop, September 2026. Photo credit: Michael Chariga.*

Programme Context

The Kenya Agroecology Hub, anchored at Manor House Agricultural Centre (MHAC), received funding from the Conservation, Food and Health Foundation to implement the project "Improving Farmer Training on AE Practices Through Collaboration and Follow-up." The project aimed to improve the impact of the one-week agroecology (AE) training for farmers through joint assessment of the AE concepts and practices that farmers were putting into practice on their farms after training, and those they were not.

The project was implemented with ten partner organizations across 10 counties, covering 42 sub-counties and involving 223 farmers from the western part of the country. The farmer workshops were done in five cohorts, with each cohort bringing approximately 40 farmers, five from each organization (the five were expected to form their own farmer group to support continued learning and sharing post-workshop). During the workshops, a baseline assessment was conducted to check where farmers were in terms of AE practices they were already implementing. After the training, farmers were given one and a half seasons to implement the AE practices, after which a joint follow-up was done by staff from partner organizations and MHAC staff.

Evaluation Purpose and Objectives

The overarching purpose of this endline evaluation was to reveal which new AE concepts and practices were being implemented, or not embraced, by farmers who attended MHAC's one-week AE workshop, and to explore, qualitatively, the barriers to adoption for practices farmers were not implementing.

While MHAC has constantly used internal learning systems, such as end-of-workshop surveys and informal feedback during daily reflections held every morning with workshop participants, this assessment aimed to find out whether there is a disconnect between MHAC's assumptions of what is really working with regard to the trainings and the actual situation on farmers' farms. This assessment therefore served to reveal whether failure to implement the fundamental AE practices and concepts covered in the workshops was due to communication gaps on the side of MHAC trainers, and/or other factors, such as a mismatch between what MHAC assumed was relevant and what farmers themselves deemed relevant and/or achievable in their context.

Methodology

This evaluation used a structured questionnaire that had a list of questions related to the practices taught in the one-week workshop. The questions were either yes or no, with yes if a farmer was doing the practice and no if otherwise. To capture barriers to adoption of these practices, the questionnaire was adapted to accommodate in-depth interviews for qualitative reasons behind a "no" response, i.e. the evaluators were to record yes or no for a practice and record the reason for not doing the practice as given by the farmer.

For the sampling of farmers, a full census approach[5] was used, with all farmers targeted for assessment rather than working from a fixed sample size. This decision was informed by foreseen uncertainties, such as the likelihood that some farmers would not be available during the assessment period, which could have reduced the number reached had a fixed sample been set in advance.

The data tool was created and piloted with 20 farmers in Trans Nzoia County, and revisions were made based on the insights from the pilot.

The evaluation team was split into two teams, with each team comprising MHAC staff and personnel from the local organization that had supported entry to farmers. The visit process was carried out in two phases: the first phase involved the farmer taking the team through his/her farm and elaborating on the practices they had implemented after the workshop, and the second phase involved sitting with the farmer to administer the

5 **Census approach.** Every targeted farmer is visited, instead of a sample of farmers.

questionnaire, asking the farmer to give reasons for the practices that were not seen (see annex for the visit protocol).

In addition to the data tool, the team had field notebooks to capture observations and any other relevant information given by farmers that would otherwise not be captured in the data tool. After each day's visit, the evaluation team met to reflect and note observations, and to share learning on contextual matters and how to handle them.

Limitations of the Evaluation

While the tool captured qualitative depth on what made farmers not implement specific practices, as well as what they were already implementing, it cannot be relied on to show the extent to which a farmer is implementing a specific practice. Unlike the TAPE tool[6], which has a scale to indicate the magnitude of a specific practice, the tool used had this deficit. This notwithstanding, the tool has rich qualitative responses that served the objectives the evaluation sought to answer.

Besides this deficit in the evaluation tool, the process did not time the prime period when farmers have things to show on their farm. Most farmers had either already harvested (for crops with short maturity), or their farms had been disrupted by weather; while there was still some trace of evidence of what they had done, some of it could not be seen. There were, therefore, other practices that farmers said they had done but for which there was no evidence; in such cases, the evaluation team had to rely on the goodwill of the farmer to be honest.

6 TAPE. The Tool for Agroecology Performance Evaluation, which uses a scale to show how extensively a practice is applied.

SECTION 2

Analysis of Findings

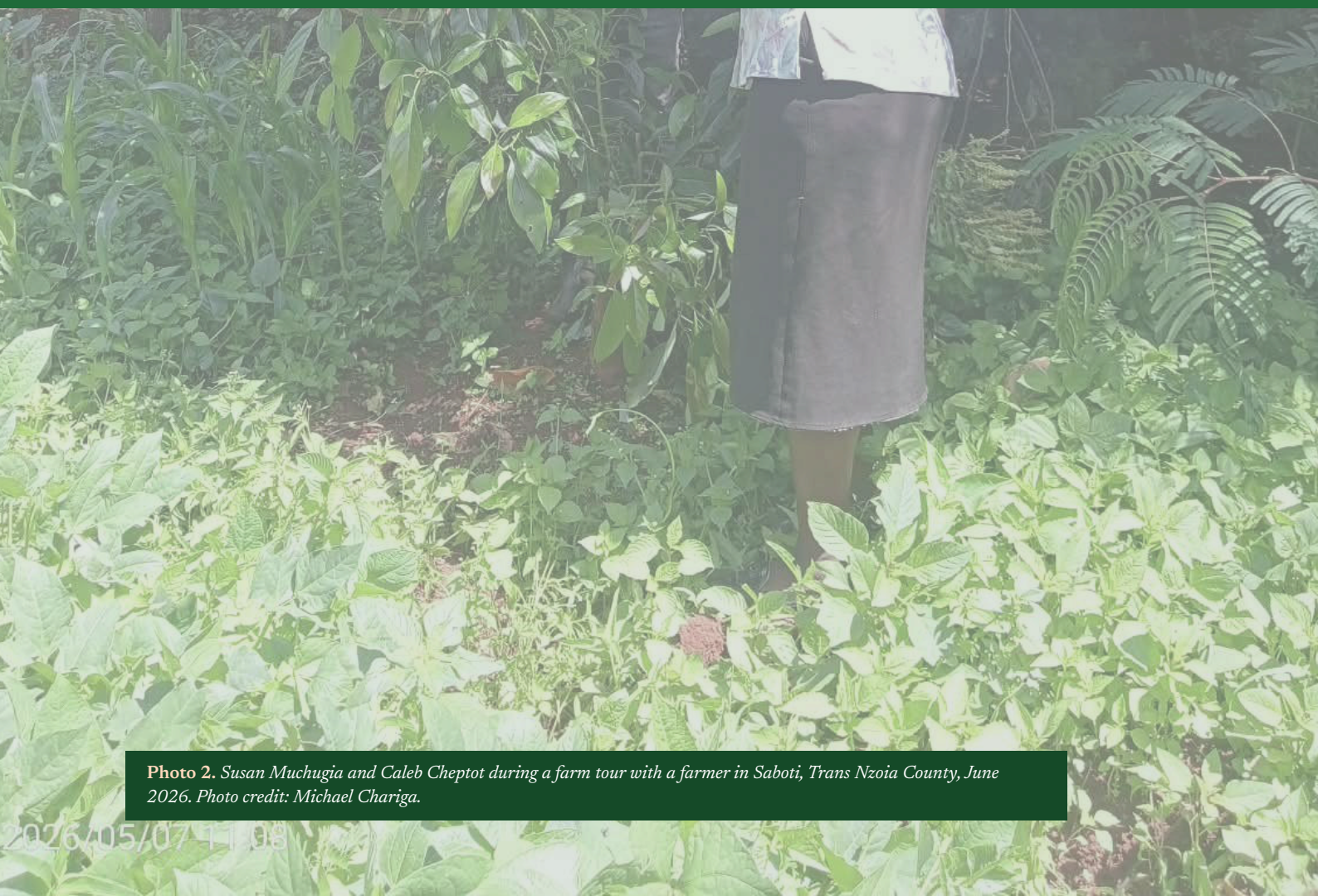


Photo 2. *Susan Muchugia and Caleb Cheptot during a farm tour with a farmer in Saboti, Trans Nzoia County, June 2026. Photo credit: Michael Chariga.*

Evaluators observations

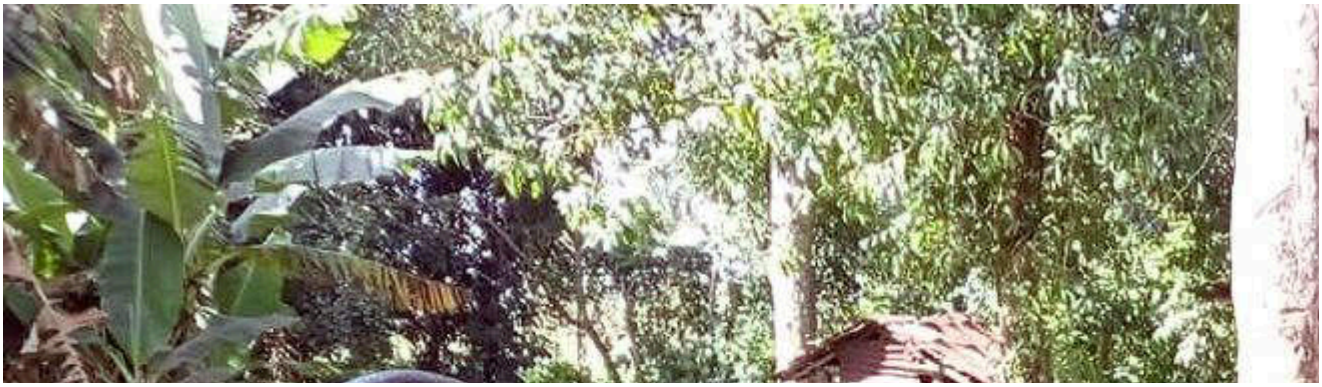


Photo 3. *Agroforestry tree cover at the home of a KALRO farmer in Kendubay, Homa Bay County, July 2026. Photo credit: Michael Chariga.*

In this section, we present insights from the evaluation team's observations, key points from the joint daily reflection sessions, and insights shared by farmers that would not otherwise have been captured by the evaluation tool. The observations are presented both generally and specific to each theme of the workshop.

The general observations are those that are not directly related to the topics taught in the workshop but provide valuable context that could help understand why farmers have not implemented some practices. The specific observations are those related to the topic(s) taught in the workshop.

General Observations:

Most first-time/New farmers farmers had a low adoption rate. The most noticeable transformation was seen with farmers who had attended workshops in other projects. Similarly, even amongst the first-time farmers, those who have high implementation levels are the ones whom the group members usually gather at their farms for demonstrating the practices/interventions, or other program meetings are held in their homesteads.

Workshop content was congested and fast-paced. While we may want farmers to learn almost everything MHAC does, there is always limited time, considering the many things in every department that would be beneficial to them. The majority of farmers said they were uncomfortable with the pace and learning too many things within a short period. This was, however, not a problem with the youth, but the majority of the older generation noted being unable to keep up with the pace, saying it was too fast for them to completely process and understand the concepts the way they would have loved to. More specifically, rotating farmers during the practicals was consistently mentioned. They requested to be given ample time to understand those practices per session. They would rather have a few sessions and understand properly than take in many things and end up with nothing to remember.

Inactivity and Overcrowding during practicals- It has also been noted with concern that the arrangement of some practical sessions hindered some farmers from understanding the practices. Approximately 8 out of every 10 farmers assessed, when given a chance to speak out/share insights, or ask questions, said that there were practices they desperately yearned to learn, but could not simply because there were so many people crowding around a trainer demonstrating how to do certain practices.

Only people at the front could actually see what was happening; those at the back were blocked. Even for those who could see, never got a chance to do the practicals on their own, but were just shown because “time was short”. Farmers noted they were not even confident enough to try them at home, saying they didn’t know how to. Additionally, all those practicals that farmers were not actively engaged in, or were just watching trainers demonstrate how to do, had little to no impact on the ground.

Need for reference materials/ Checklist. The majority of farmers noted that they could not remember the processes of how to do some practicals, or entirely forgot about them, and requested that future workshops should consider providing reference materials to help them remember those processes.

Authority on the use of land. Some farmers may be disadvantaged since they have no say in the land; they are not the owners. In other cases, a farmer may be practising/implementing on land under a lease, limiting what they can actually adopt /the choice of interventions.

Lack of seeds. Some seeds are not available in local markets and can only be found at MHAC. The " new/ alternative crops," for instance, the likes of buckwheat, Quinoa, and chia seeds, are hard to find elsewhere. For these, farmers said they didn't have them on their farms because they lacked seed. Ideally, when they attended the workshop, they were not able to purchase seeds. Most of them noted that when leaving their homes, they were unaware there were things they needed to purchase, so they were unprepared financially. They left the workshop wishing they had enough money to buy those seeds. Some of them explained that they were able to purchase a few things, while others were totally unable to purchase anything due to financial constraints.

Seed Viability for germination. Some farmers were happy with the seeds they got, which had good germination and were growing well. In fact, some of those satisfied had produced seeds they harvested to display to us. However, across the regions, the issue of some seeds not germinating was common. While there could be a number of factors that could inhibit seed germination, MHAC may need to revisit the storage procedures to make sure they don't overstore seed to the point of losing viability. Most of the seeds mentioned include tephrosia[7], soyabeans, and Number Nane seeds.

Still concerning tephrosia seeds, the team has severally encountered farmers having both purple-flowered and white-flowered ones, both germinating out of seeds they got from MHAC. It indicates that the inactive and active Tephrosia species got mixed up during harvesting or post-harvest handling.

Youth Implementation Rate is still low: A few youths have done something significant worth noting. Despite the high potential we have seen in youths who have a level of implementation, the majority are constrained in terms of access to land. A few who have had some level of access to land have had some projects running, but have had limited access to financial and material resources.

Youth Availability- Farmers missing on some days. A consistent observation is that farmers who missed out on specific days during the workshop have a gap evident even in the implementation. A clear gap exists because they have no clues about what those specific practices are. For instance, there are some organizations where farmers arrived on Tuesday during the day and found the workshop ongoing. Those farmers missed out on soil health, including the theory, the practical part, and also missed the morning recap. The knowledge gap for soil health is undeniable.

This also applies to those farmers who left earlier due to unavoidable circumstances and emergencies, and missed out on specific days

*Unpredictable Weather Patterns***:** Across the regions, farmers have cited experiencing weather challenges. There has been erratic, totally unpredictable rainfall, and farmers can no longer schedule their farming calendar and activities around it. When rains are expected to be continuous, they fail; farmers already planted, it rains for a short period, and disappears, leaving crops on farms struggling for moisture. When it finally rains, it's not the right time to benefit crops; the critical growth stages that crops require moisture have elapsed; they have already withered.

Farmer follow-ups and implementation levels: Remarkable progress was evident among farmers who have been visited. After training, some partner organizations engaged their farmers with other projects/programs. These have received multiple follow-up visits from extension officers, which increased farmers' motivation to implement more. Most of these farmers have became lead farmers, recruiting and training other farmers in their communities. A wider, undeniable gap exists between those frequently followed up and those who had never been visited since training.

Qualitative and Quantitative Data Collected

This section focuses on qualitative and quantitative data collected using the data collection tool for the five days training areas. Each day opens with the result for all of its practices, then walks through the findings, and ends with recommendations. Because the days build on one another, each section starts by linking back to what the previous day showed. The last section pulls the five days together.

7 **Tephrosia.** A legume shrub whose leaves and seed are used in botanical sprays against pests.

Day	Baseline surveyed	Endline reached	Has both baseline and endline data	Number of Practices	Rose	Fell
Day 1: Soil health	216	141	131	14	11	3
Day 2: Pest management	220	141	136	8	8	0
Day 3: Nutrition and value addition	218	141	134	14	9	5
Day 4: Livestock	220	141	134	12	9	3
Day 5: Climate adaptation	221	141	133	15	11	4

All comparisons use matched farmers[8], which is about 60 to 62% of each day's baseline sample. Percentages are shares of matched farmers who answered each question, so the denominator varies slightly from practice to practice. Three points help in reading the numbers:

- "Not observed" is not always "stopped". Where a practice was not seen or reported at the visit, the farmer may simply not have been doing it that season. This matters most for seasonal crops such as finger millet and for botanical remedies that are only prepared when needed.
- Some questions are worded in reverse. On Day 2, rotation (Q1) and renewing seed and planting materials (Q7, Q8) are coded so that a "yes" means the desired practice. The report describes them in the desired direction.
- Evidence quality is high. Between 95% and 98% of endline "practising" responses on Days 2 to 5 were confirmed by direct observation, compared with 84% on Day 1.

8 **Matched farmers.** Farmers with both a baseline and an endline record, so that their answers can be compared.

DAY 1

Soil Health



Photo 4. *Mulched cone garden established by a DIG farmer in Ndhiwa, Homa Bay County, July 2026. Photo credit: Michael Chariga.*

Farmers were trained on 14 soil health practices and assessed at baseline and again at endline. At baseline, 216 farmers were surveyed and 212 had complete soil health entries. At endline, 141 farmers were reached. All comparisons in this section use the 131 farmers with complete baseline and endline records, about 61% of the baseline sample. Percentages are shares of matched farmers who answered each question (120 to 131 depending on the practice).

Overall results for soil health practices

Figure 1 shows the change for every practice. Composting and liquid fertiliser practices rose sharply. The three biomass practices at the top of the "decline" list are the exception, and they are discussed first in the findings.

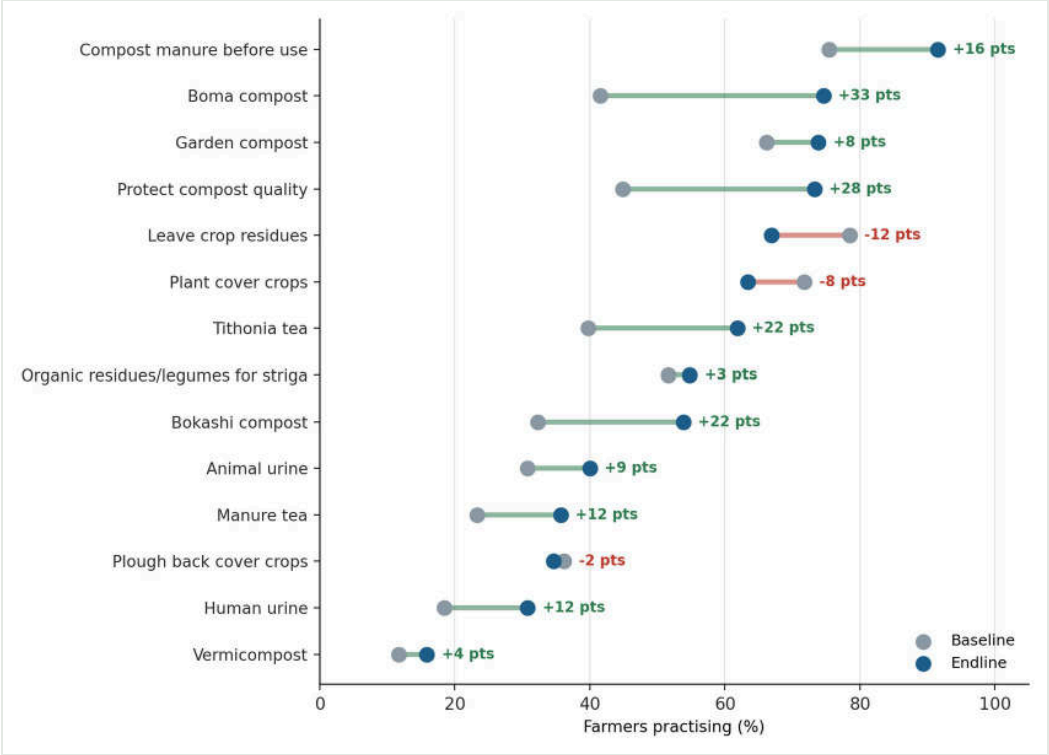


Figure 1. Share of matched farmers practising each practice at baseline and endline (points show the change). Source: CFHF baseline-endline matched data

Behind the net change, many farmers moved in both directions. Figure 2 splits each practice into farmers who continued, newly took it up, stopped, or never practised. For residues and cover crops[9], the net decline hides real churn: 33 farmers stopped leaving residues while 18 started, and 34 stopped planting cover crops while 23 started. For vermicompost, human urine and bokashi[10], the large grey section shows that most non-adoption is farmers who never started.

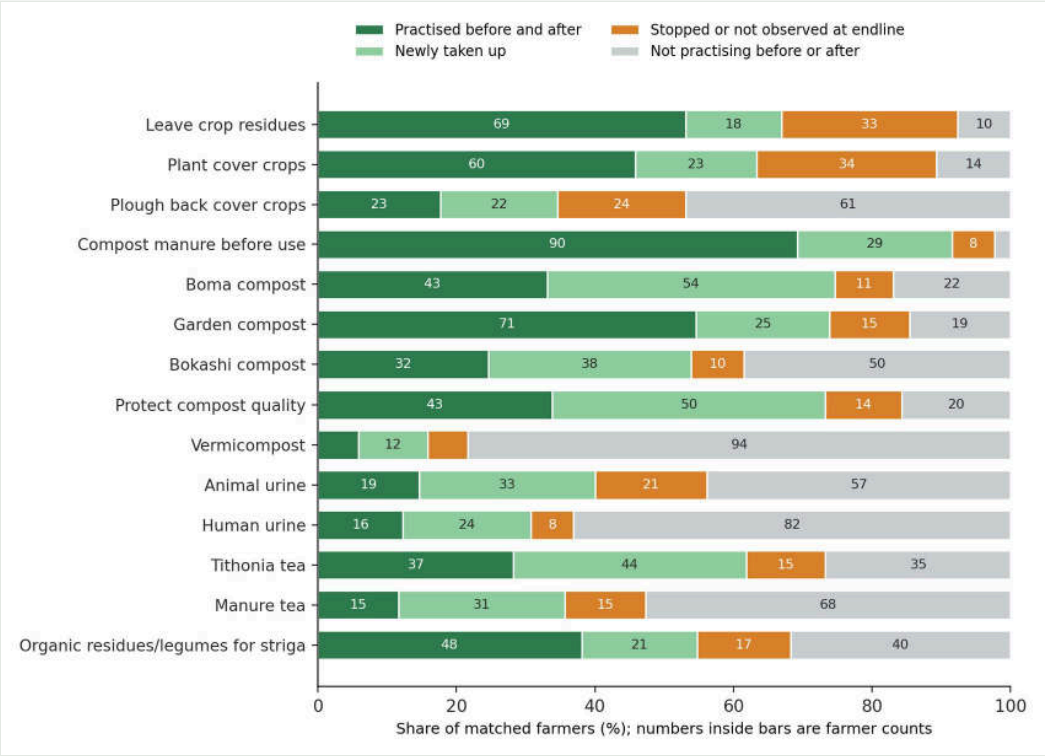


Figure 2. What happened to each farmer between baseline and endline. Numbers inside bars are farmer counts.

Soil Health Findings:

Finding 1: Practices that compete with food, feed and seed lost ground

Crop residues Of 131 matched farmers, 102 left crop residues in the field at baseline and 87 did so at endline, a net fall of 15 farmers. Thirty-three farmers stopped and 18 started. Of the 33 who stopped, 19 diverted residues to livestock feed and 9 to compost or manure making (Figure 3, left). Residues diverted to compost still return to the farm, so part of this decline is a change in route rather than a loss for the soil.

The question may also understate the practice. It asked whether farmers leave residues to feed termites. Farmers who do leave residues mostly do so for mulching and moisture retention, or for soil fertility through decomposition or incorporation. None described it as termite control. Some farmers, especially youth and women, also had no authority over land use and could not persuade a spouse or parent to adopt the practice.

IMPLICATION

The decline is real but smaller than it looks, and part of it reflects how we framed the question. Termites are not the only organisms that feed on residues, and the alternative uses farmers gave all support long-term soil improvement.

ACTION

Rephrase the question around any return of residues to the soil (for example "food for macrofauna"), or drop the termite wording. Move termite management to Day 2 (pest management).

9 Cover crops and green manure. Crops grown mainly to protect and feed the soil, sometimes ploughed back before harvest.
10 Bokashi. Compost made by fermenting organic material with ingredients such as ash, biochar or molasses. It is ready faster than ordinary compost.

Cover crops and green manure Planting cover crops fell from 94 to 83 farmers, a net fall of 11 (34 stopped, 23 started). Ploughing cover crops back stayed low, from 47 to 45 farmers, with 61 farmers never practising it (Figure 3, right). Field notes explain why: cover crops are usually grown for seed rather than incorporated, and farmers see ploughing them under before harvest as "wasting food".

Farmers also know few species, mostly common beans. Where farmers did incorporate lablab or soya, it was mainly by research groups such as those in Trans Nzoia (MHAC), or because the crop had failed in a drought. Some farmers also treat burying weeds during weeding as the same as green manure.

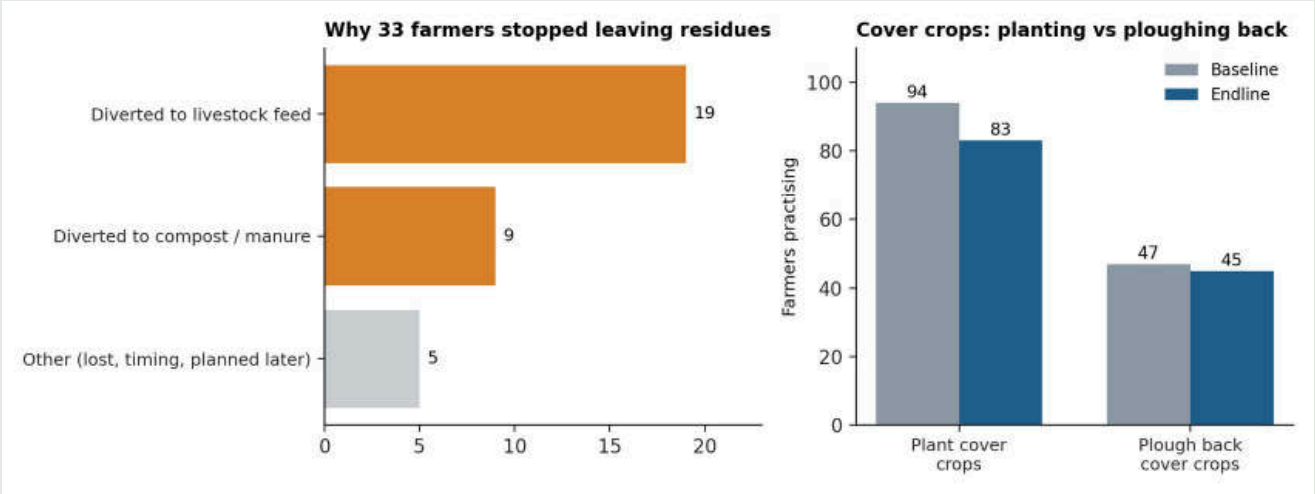


Figure 3. Left: reasons given by the 33 farmers who stopped leaving crop residues. Right: farmers planting and ploughing back cover crops, baseline and endline.

One caution on the evidence. In the coded survey reasons, only five farmers explicitly cited seed or food production. Among the 85 farmers not ploughing back at endline, 41% gave no specific reason or said they planned to do it later.

IMPLICATION

The barrier is competition with food and seed, not awareness. A practice that costs farmers food will not be adopted, however well it is taught.

ACTIONS

- → Give farmers a list of cover crop and green manure species, so that common beans are not the only example.
- → Promote a non-food option. Tithonia[11] is not nitrogen-fixing, but it has nitrogen content similar to leguminous plants such as calliandra and decomposes quickly, so it suits short-term nutrient release. KALRO soil health restoration results showed good growth where tithonia biomass was incorporated and a tithonia tea top-dress was applied. Farmers already relate to it, and tithonia tea uptake rose from 52 to 81 farmers.
- → Encourage a mix of short-term fertility (tithonia, lablab) and long-term soil health that builds carbon, such as incorporating maize stover.
- → Teach the difference between green manure crops and common weeds, and when to incorporate them.

Finding 2: Composting is the strongest area of uptake

Farmers have embraced composting. Composting manure before use rose from 98 to 119 farmers (75% to 92%). Boma compost[12] rose from 54 to 97 farmers, garden compost from 86 to 96, and protecting compost quality from 57 to 93. During assessment, many homesteads had compost piles, either boma compost, garden compost, bokashi, or a combination.

11 Tithonia. A bushy plant whose leaves are used as green manure or to make liquid fertiliser (tithonia tea).
12 Boma compost. Compost made from livestock manure and bedding collected in the animal enclosure (boma).

Boma compost was the most common because materials are readily available and livestock make it easier to prepare. Farmers without livestock mostly had garden compost. A further group made a hybrid pile that combined dry material, green material, animal waste and kitchen refuse, with bokashi ingredients such as ash, biochar or molasses added, then turned occasionally until ready.

IMPLICATION

Low-cost practices built on materials farmers already have are taken up quickly. They are the strongest entry point for the rest of the Day 1 package.

ACTION

Link the residue message to composting, so that residues diverted to compost are recognised as a valid way to return biomass to the soil.

Finding 3: Bokashi grew, but input access caps it

Bokashi rose from 42 to 70 farmers (32% to 54%). Thirty-eight farmers took it up, 10 stopped, and 50 never practised it. Among the 60 farmers not practising at endline, 47% cited access to inputs, 22% cost and 13% labour or time (see Figure 4 in Section 4). Molasses and other purchased ingredients, or ingredients sourced from far away, are the main problem.

Farmers who use bokashi value that it is ready quickly, without the months of waiting that boma and garden compost need, and some are willing to buy the materials. FIPs Africa redesigned the recipe with cheaper alternative ingredients, which made it affordable to more farmers, although quality may differ. Bokashi also works well against striga[13], which has raised demand among neighbours, and some FIPs youth now sell it as a business.

IMPLICATION

Uptake is held back by supply of materials, not by a training gap.

ACTIONS

- → Prepare two bokashi piles in each practical session, one with purchased ingredients and one with alternatives, side by side. Farmers tend to treat what the trainer does as the correct way, so a listed alternative is less convincing than a demonstrated one. Tell them that nutrient quality may differ.
- → Help farmer organisations source molasses from affordable suppliers.
- → Support the youth who already sell bokashi, since this also widens access for other farmers.

Finding 4: Vermicompost is the weakest practice

Vermicompost is the lowest practice at both baseline (14 farmers) and endline (19 farmers), and 94 of the 120 farmers with data never practised it. Among the 101 farmers not practising at endline, 41% cited access to materials (worms and construction materials), 21% a knowledge or skill gap and 8% cost. Farmers who tried it reported that chickens and other birds ate the worms, and some had security concerns. Many did not know where to get worms, and those who did said they were expensive.

IMPLICATION

Farmers see vermicompost as expensive and risky. No amount of repeated classroom training will help unless access to worms and materials is solved.

13 Striga. A parasitic weed that attacks cereal crops such as maize and sorghum.

ACTIONS

- → Arrange starter worms at farmer-friendly prices, or support a community vermicomposting site for farmers who live close together.
- → Run practical sessions where farmers build the structure themselves, with trainers facilitating, and show cheaper improvised materials.
- → Include protection against birds in the demonstrated design.

Finding 5: Liquid fertilisers are growing, but urine practices hit structural limits

Tithonia tea rose from 52 to 81 farmers (40% to 62%), the largest gain outside composting. Manure tea rose from 30 to 46 farmers, with barriers split between a knowledge gap (36% of non-practising farmers) and no specific reason or plans for later (43%).

Animal urine rose from 40 to 52 farmers (31% to 40%), but 57 farmers never practised it and 21 stopped. Of the 78 farmers not practising at endline, 97% cited having no livestock or too few. A few said they did not know how to collect it.

Human urine rose from 24 to 40 farmers (19% to 31%), still from a low base, and 82 farmers never practised it. Among the 90 farmers not practising at endline, 53% cited social, cultural or fear-related reasons, and another 29% gave no specific reason or planned to try later. This is a stigma and safety concern rather than a knowledge gap, and a refresher training is unlikely to change it quickly.

ACTIONS

- → Animal urine: continue training and use farmers who already collect urine to lead a demonstration on how to do it. The livestock constraint itself cannot be fixed through training.
- → Human urine: set up a working demonstration site at MHAC so farmers can see the practice, and invite practising farmers to share their experience. Running it as a FRN research trial, as suggested, would add credibility.

Finding 6: Striga control is practised, but for other reasons

Use of organic residues or legumes to manage striga changed little, from 65 to 69 farmers (52% to 55%). Of the 57 farmers not practising it at endline, 58% were using the inputs for another purpose and 39% gave other reasons, mostly that striga is not a problem in their area. Striga is region-specific and is less of a problem in western and north rift areas. Farmers are planting legumes and using organic residues, but not with striga in mind, and these are among the low-hanging fruits that need no purchased materials.

ACTION

As with residues, ask separately about the practice and its purpose, and report results by region.

What holds soil health practices back

Figure 4 compares the reasons farmers gave for not practising at endline. The pattern shows that different practices need different responses, which is why the recommendations below are grouped by barrier rather than by training day.

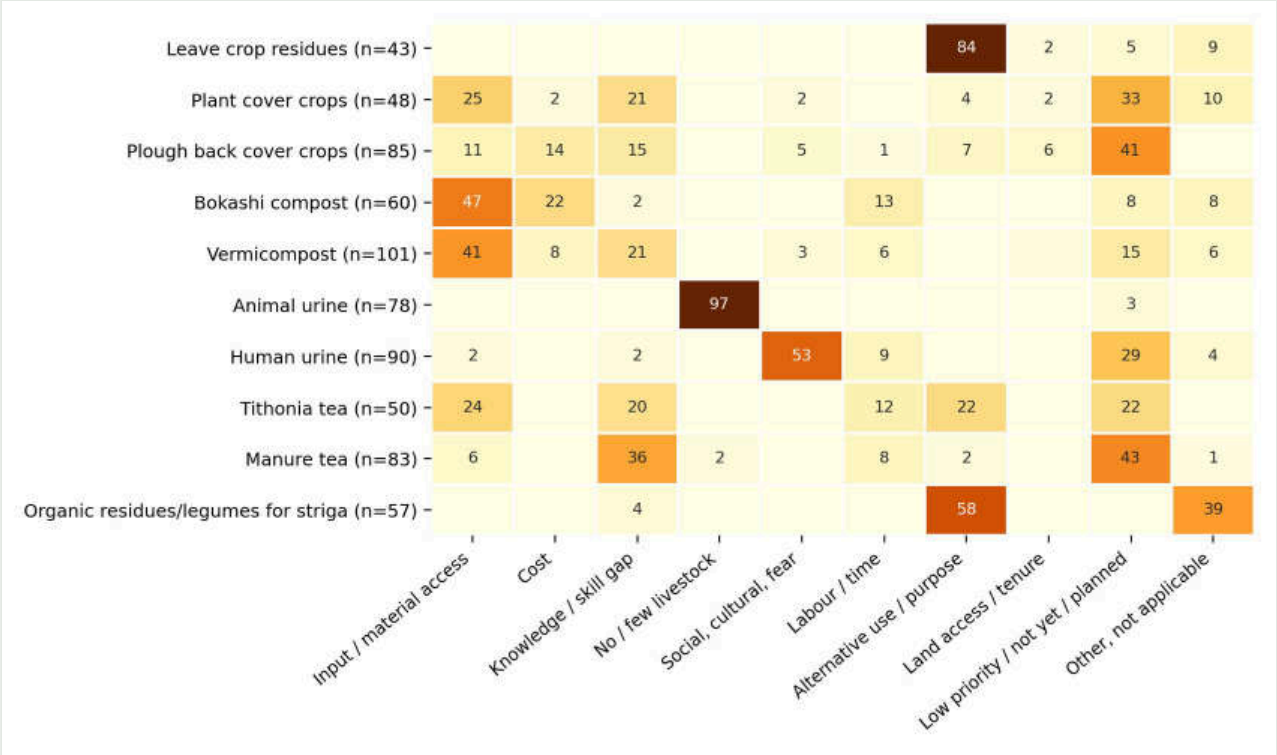


Figure 4. Reasons given by farmers not practising each practice at endline (stopped or never adopted), as a percentage of those farmers. Selected practices with the most non-practising farmers.

Barrier	Where it shows up	What response fits
Question framing	Crop residues, striga	Rewrite the indicator; separate practice from purpose
Competition with food, feed, seed	Residues, cover crops, plough-back	Non-food alternatives such as tithonia; wider species list
Input and material access	Bokashi, vermicompost	Demonstrate alternatives; arrange supply of molasses and worms
Livestock numbers	Animal urine, boma compost	Continue training; offer garden compost alternatives
Social and cultural norms	Human urine	Visible demonstration site; peer testimony; research trial
Decision-making power	Youth and women on family land	Include household decision-makers in invitations

DAY 2

Pest management



Photo 5. *Striga infestation on a farm in Matayos, Busia County, June 2026. Photo credit: Michael Chariga.*

Day 1 ended with a pattern: practices that use what the farmer already has were taken up, and practices that need bought inputs stalled. Day 2 follows the same pattern, with a stronger result. All eight practices rose, and the largest gains came from practices that rely on plants and knowledge rather than purchases. Figure 5 shows where each practice started and finished.

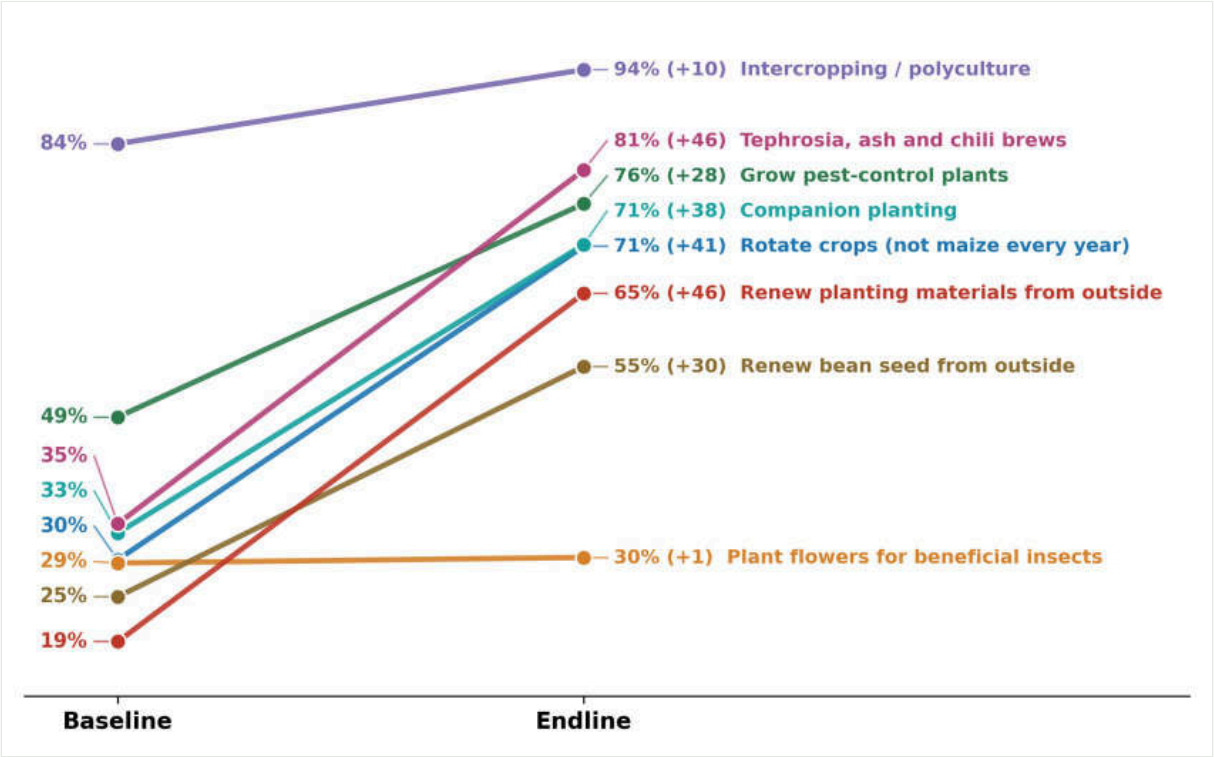


Figure 5. Share of matched farmers practising each Day 2 practice at baseline and endline. The orange line is the one practice that barely moved.

Finding 1: Crop rotation is spreading, but farmers rotate within the same crop family

Rotation (not planting maize in the same field every year) rose from 40 to 95 farmers (30% to 71%). Sixty-four farmers took it up and 31 continued.

The quality of rotation is a different matter. Field notes show that most farmers understand what rotation means but follow a shallow sequence. Many simply alternate maize and beans, or move from maize to sorghum or millet, which are in the same family. In Trans Nzoia, when

asked what grew before maize, most farmers said beans and said beans would follow again. Of the 39 farmers not rotating at endline, 16 plant maize every year (intercropped or as a monocrop), 8 prioritise maize for food security or feed, 6 cited a knowledge gap and 3 a small land size (Figure 6).

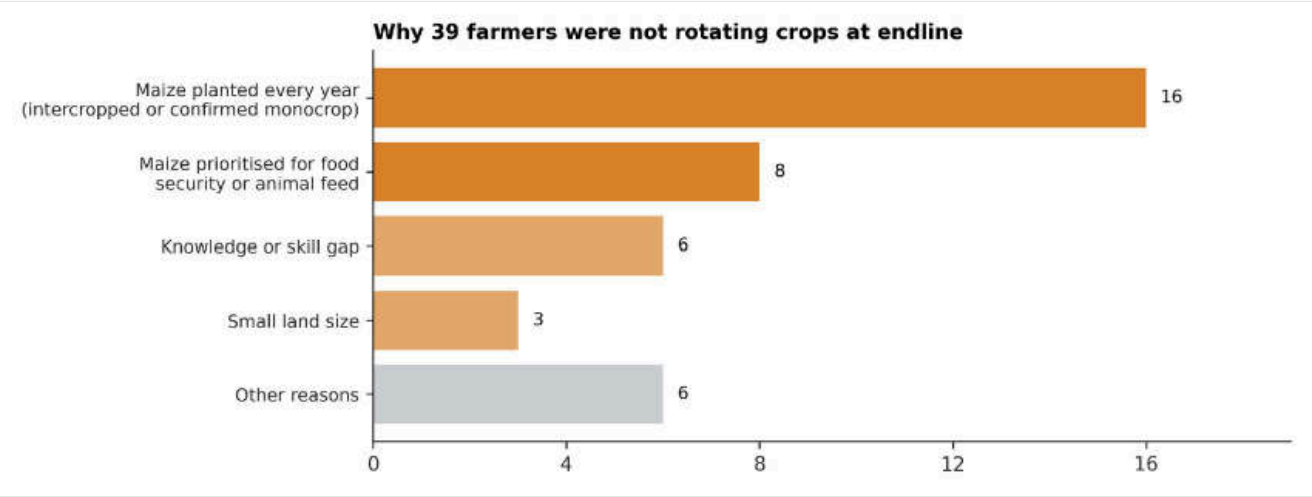


Figure 6. Reasons given by the 39 farmers not rotating crops at endline.

IMPLICATION

This is more of a knowledge gap than a resistance problem. Farmers are receptive, so giving them the missing information could let them rotate properly.

ACTIONS

- → Tell farmers which crops are related and should not follow each other, like "sisters and brothers", and list those crops.
- → Use a simple picture showing crops in the same family and light and heavy feeders, with arrows showing what should follow what. Review the pictures farmers already know and correct them where needed.
- → Provide a simple three-year rotation schedule for farmers with small land divided into two, three or four plots, listing the crop options for each plot over three years. It should be much simpler than the MHAC model.

Finding 2: Botanical and companion practices grew fastest, with few barriers

Tephrosia, ash brew and chili sprays rose from 47 to 110 farmers (35% to 81%), the largest gain on Day 2 along with renewing planting materials. Companion planting rose from 45 to 96 (33% to 71%), and growing pest-control plants such as tephrosia, tithonia and aloe vera rose from 66 to 104 (49% to 77%).

Barriers were small and varied. Among the 26 farmers not using botanical brews, 6 preferred purchased chemical pesticides, 4 had a knowledge gap and 3 saw no pest pressure. Companion planting was held back mostly by drought losses and forgotten training (8 farmers each). In 9 of the 32 cases where pest-control plants were not counted, the plant was present but used for another purpose or not deliberately for pest control.

IMPLICATION

These practices work because they use plants and ingredients farmers already have, and because pests are a visible problem. They are the strongest part of the package.

ACTION

Keep teaching them as they are. Where a pest-control plant is already on the farm for another reason, show farmers how to use it for pest management.

Finding 3: Flowers for beneficial insects show high churn and one clear barrier

Planting flowers is the weakest Day 2 practice. The net figure barely changed (40 to 41 farmers), but that hides a lot of movement: 14 farmers continued, 27 started, 26 stopped and 69 never practised it (Figure 7, left).Of the 95 farmers not growing flowers at endline, 62 (65%) cited access to seeds or materials, and another 16 (17%) lost the crop to drought or pests (Figure 7, right). Only 3 cited a knowledge gap.

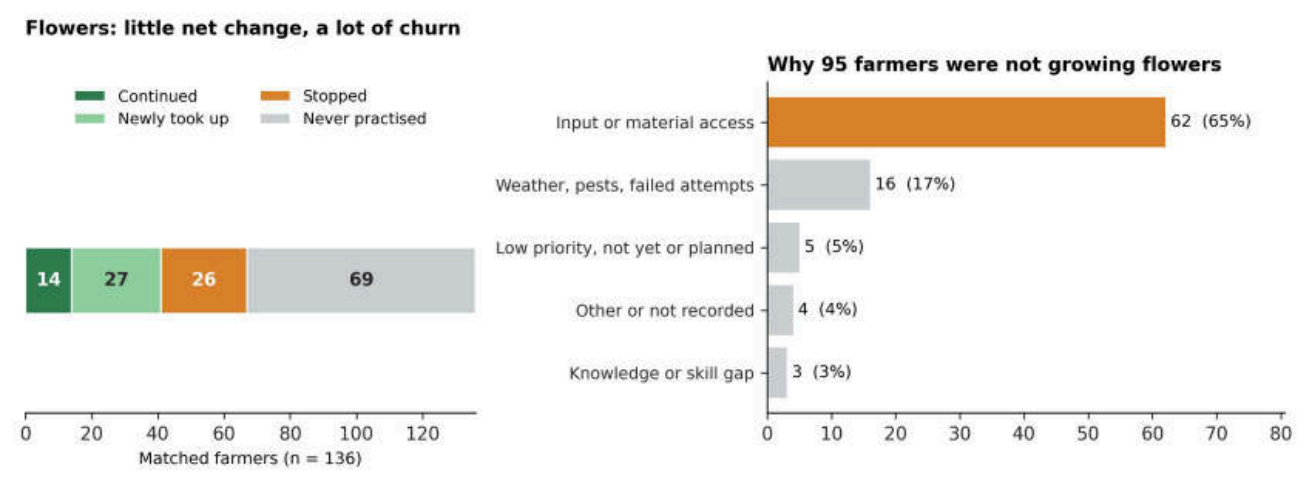


Figure 7. Left: what happened to each farmer for flowers. Right: reasons given by farmers not growing flowers at endline.

IMPLICATION

Without a fix for seed access, future rounds will show the same result.

ACTION

Prepare starter packets of flower seed before training so that farmers leave with seed in hand. Even giving packets to a team leader to share would help. Choosing hardy species would also reduce drought losses.

Finding 4: Intercropping is common, but it is done for soil fertility rather than pest control

Intercropping or polyculture was already the most common Day 2 practice and rose further, from 113 to 126 farmers (84% to 94%). One hundred and eight farmers continued and 18 took it up after training. Most farmers intercrop maize with beans, and sometimes soya or groundnut. The way farmers described it suggests they do this to improve soil fertility, not to manage pests, even though it serves both purposes.

The question combines intercropping and polyculture, so the high figure hides the fact that true polyculture is rare. Companion planting, which is more intentional, sits at 71%.

IMPLICATION

Intercropping will stay high regardless of training. The training effect is the 18 new adopters, and the survey cannot yet show whether polyculture has changed at all.

ACTIONS

→ Continue to teach intercropping and polyculture, and explain which crops suit pest management, not only beans.

→ Split the question into two separate questions, one for intercropping and one for polyculture, so that each can be measured.

Finding 5: Renewing seed and planting materials is rising, but farmers have reasons to keep their own

Sourcing bean seed from outside the farm rose from 34 to 75 farmers (25% to 55%), and sourcing planting materials such as banana, cassava and sweet potato rose from 26 to 88 (19% to 65%). The second was one of the biggest gains of the day (Figure 8).

Even so, 61 farmers still replant their own bean seed and 48 their own planting materials. Most did not give a detail, but among those who did, saving cost was the main reason (13 and 8 farmers), followed by trust in their own seed (4 and 5). Field notes add that many farmers have bought seed that failed to germinate, that they value their indigenous seed and believe it yields more, and that purchased material is expensive while their own is free.

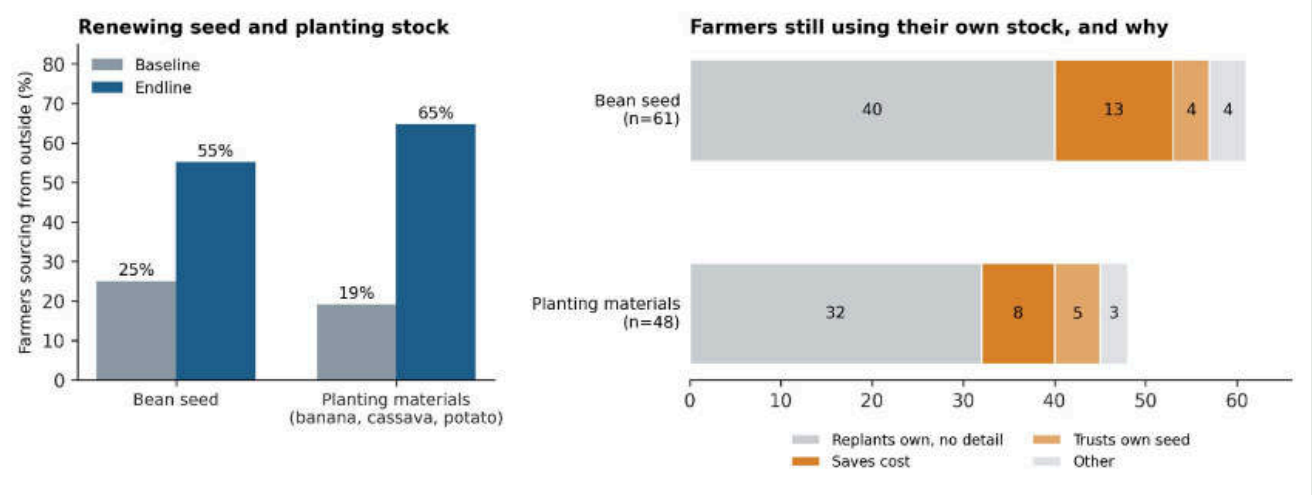


Figure 8. Left: farmers sourcing seed and planting materials from outside, baseline and endline. Right: reasons given by farmers still using their own stock.

IMPLICATION

Keeping own seed is a reasonable response to poor seed quality and cost, not a lack of understanding. Training should focus on planting disease-free planting materials.

ACTION

Present renewal as occasional rather than all-or-nothing, and focus it on vegetative materials where disease builds up. Help farmers find seed sources they trust and teach a simple germination test before planting.

DAY 3

Nutrition and value addition



Photo 6. *Bananas on the plot of an RFRN farmer in Nyatike, Migori County, July 2026. Photo credit: Michael Chariga.*

Day 2 ended with seed access as the main brake on flowers. Day 3 meets the same wall with new crops, and adds a second problem: after farmers produce a value-added product, the later steps of packaging, marketing and record keeping fall away. Day 3 has the widest spread of results. Nine of 14 practices rose and five fell (Figure 9).

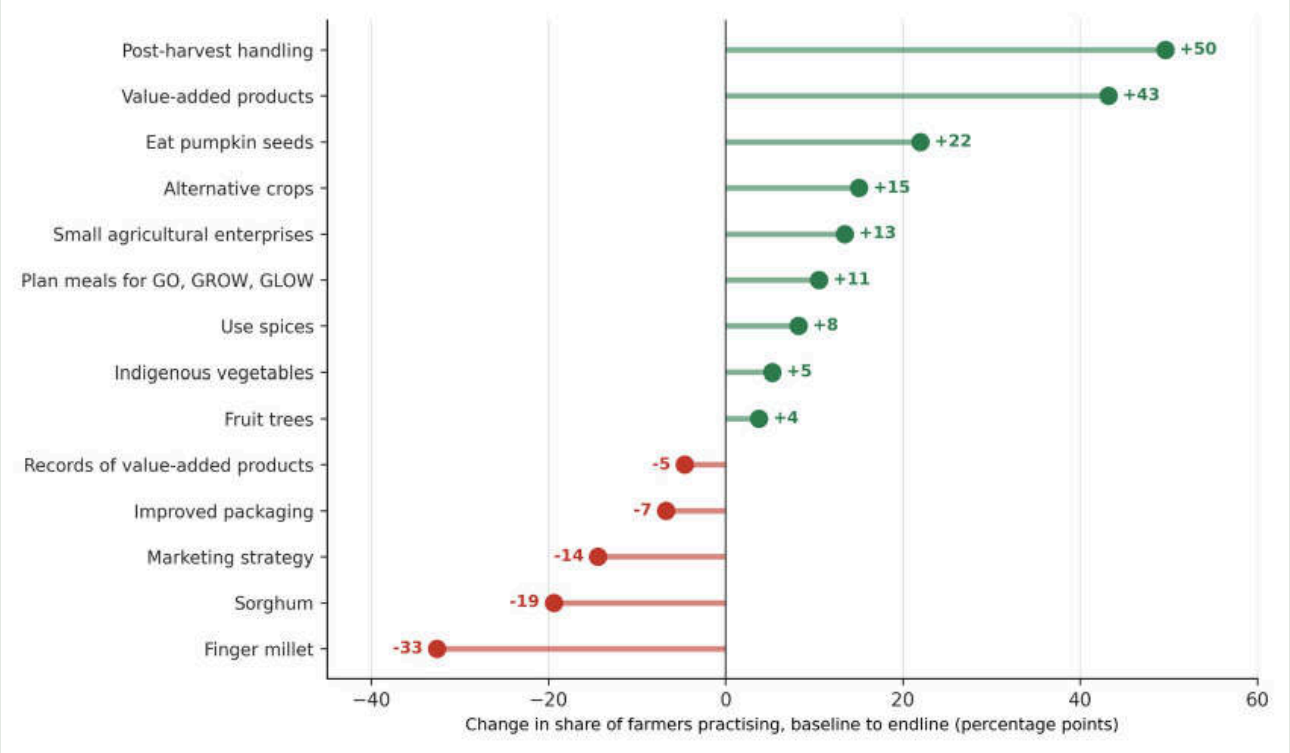


Figure 9. Change in the share of farmers practising each Day 3 practice, in percentage points.

Finding 1: Food-based practices are already well established

Five practices started high and rose further. Planning meals around GO, GROW and GLOW[14] foods rose from 81% to 92% (25 new farmers), indigenous vegetables from 93% to 99%, fruit trees from 92% to 96%, spices from 66% to 74% (26 new) and small agricultural enterprises from 82% to 96% (23 new). At some point, almost every farmer had considered family nutrition.

Gaps are small and mostly practical. Farmers missing one nutrient group were usually missing it because the crop was not available at the visit or had dried in the season's drought. A more important point is that plant protein sources are limited: most farmers have one, two or at most three. Fruit trees were mostly established long ago, with only a small group planting seedlings after training, and the few without them have low diversity or new farms. Some farmers prefer to buy spices rather than grow them, but the question only asked about using them. The few farmers without an enterprise are deliberately growing for household use.

IMPLICATION

These are common, doable practices with no dominant barrier to fix.

ACTIONS

- → Prepare a list of legumes and confirm which ones farmers know locally, to widen protein sources.
- → Link nutrition to the Day 5 message on drought tolerant crops, since drought is what removes nutrient groups from the farm.
- → Continue promoting indigenous vegetables.

Finding 2: Post-harvest handling had the largest gain of the day

On-farm food preservation and post-harvest handling rose from 52 to 115 farmers (41% to 91%), with 67 farmers starting after training. Field notes show that 80 to 90% of those practising use hermetic storage bags[15], often with drying and ash. Use of hermetic bags, however, was not a practice taught during the workshop suggesting that farmers got this from elsewhere. Of the 12 farmers not practising, 3 used chemical treatment and 2 did not have enough harvest to store.

ACTION

Continue to emphasise post-harvest handling and seed storage to reduce losses.

Finding 3: New crops and pumpkin seeds are held back by seed access and competing use

Alternative crops such as quinoa, buckwheat and amaranth rose from 33 to 53 farmers (25% to 40%). Thirty-eight farmers started, but 62 never did and 18 stopped. Of the 80 farmers not growing them, 56 (70%) cited seed access, 11 a knowledge gap and 7 drought (Figure 10, left). Field notes describe the knowledge side vividly: some farmers still had seed as bought, a season and a half later, because they did not know when to plant buckwheat. Others grew chia and buckwheat to ripeness and did not realise it was ready, so the seed shattered and was lost.

Eating pumpkin seeds rose from 49 to 78 farmers (37% to 59%), with 44 starting. Among the 54 farmers not eating them, 26 (48%) keep the seed for planting (Figure 10, right). Farmers who stopped had eaten pumpkin seeds before, but weather left them too little to eat, so what they had went to planting.

14 GO, GROW and GLOW foods. Food groups used in nutrition education: GO foods give energy, GROW foods help the body grow and repair, GLOW foods protect health.

15 Hermetic storage bags. Airtight bags that keep out air and pests to protect stored grain and seed.

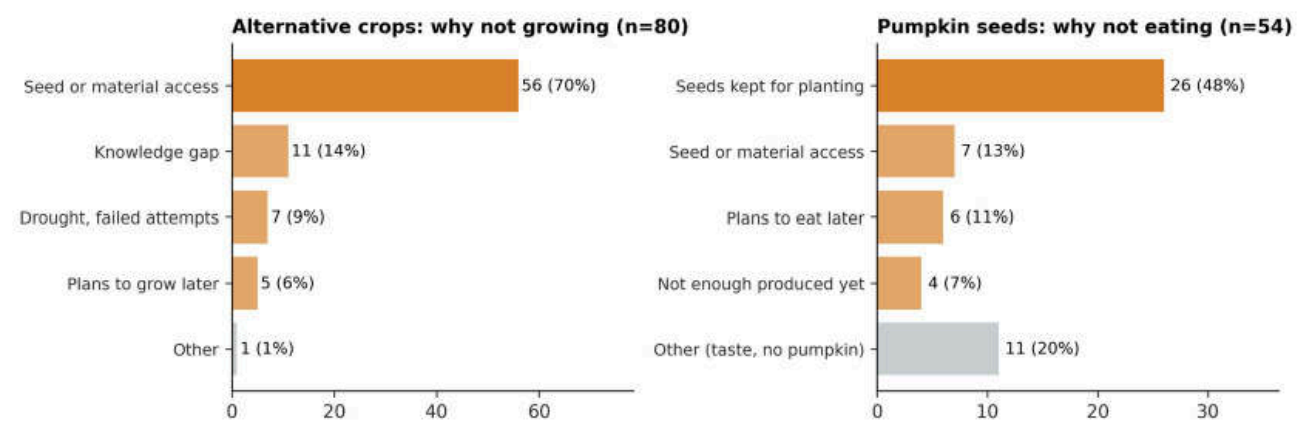


Figure 10. Left: reasons farmers were not growing alternative crops. Right: reasons farmers were not eating pumpkin seeds.

IMPLICATION

Farmers welcome the new crops, but seed access and crop management block them. Pumpkin seed is a competing-use case, not a training gap: the same product serves planting and eating.

ACTIONS

- → Find a way to provide farmers with seeds, in the same way as the flower seed packets on Day 2.
- → During farm tours,emphasis should be made to farmers on management practices of the new crops. Farmers should learn when to plant and harvest, and then learn to prepare them.
- → Keep encouraging farmers to produce enough pumpkin for both planting and eating.

Finding 4: Finger millet and sorghum are declining, mainly for environmental reasons

These are the two largest declines of the day. Finger millet fell from 79 to 36 farmers (60% to 27%) and sorghum from 91 to 66 (71% to 51%). In both, many farmers show as "not observed" (54 and 36), which does not always mean a farmer has stopped: some grew the crop in earlier seasons but not this one. Across the 159 non-practising responses, 38 cited weather, birds or pests, 25 said they had grown it in earlier seasons, 20 a small land size and 12 a dislike or poor perceived performance (Figure 11). Birds are a particular threat to finger millet.

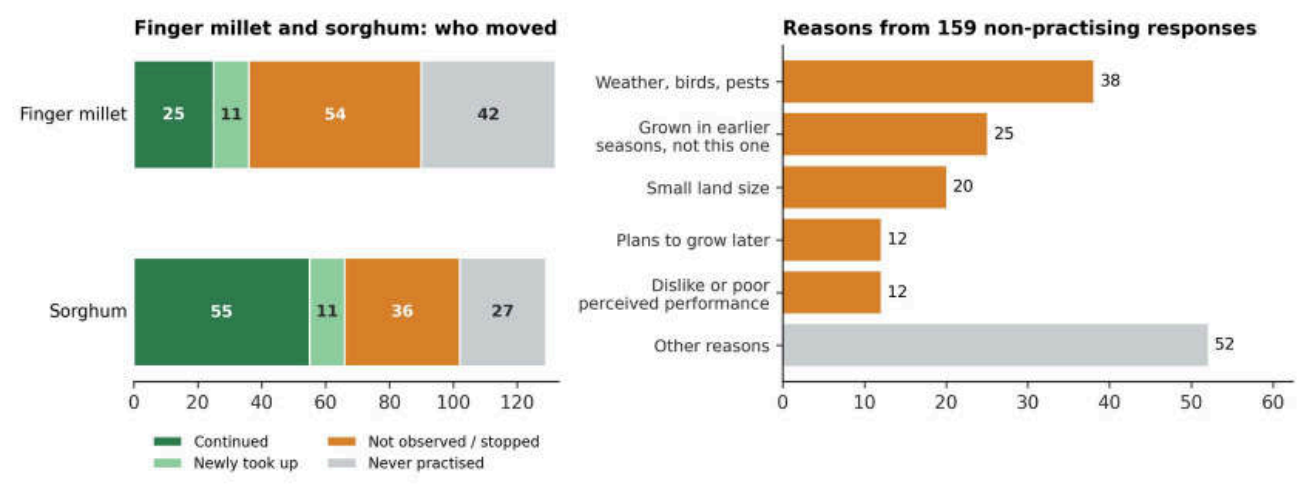


Figure 11. Left: what happened to each farmer for finger millet and sorghum. Right: reasons given across both crops.

IMPLICATION

This is not a training gap. It is environmental vulnerability, bird damage and limited land. The size of the drop does suggest farmers are moving away from these crops.

This sits uneasily with Day 5, where growing drought tolerant crops rose from 79% to 93%. Sorghum and finger millet are classic drought tolerant crops, so a decline in both works against the climate message.

ACTIONS

→ Discuss bird control, such as timing and group scaring, and variety choice with farmers.

→ Change the survey question to ask whether the crop was grown in the last two seasons, so that seasonal gaps are not counted as discontinuation.

Finding 5: Farmers learned to make value-added products, but later steps are missing

Producing value-added products rose from 18 to 75 farmers (14% to 57%), with 67 starting after training. Yet the three steps that follow all fell: packaging from 17 to 8, marketing or awareness from 38 to 19 and record keeping from 34 to 28 (Figure 12).

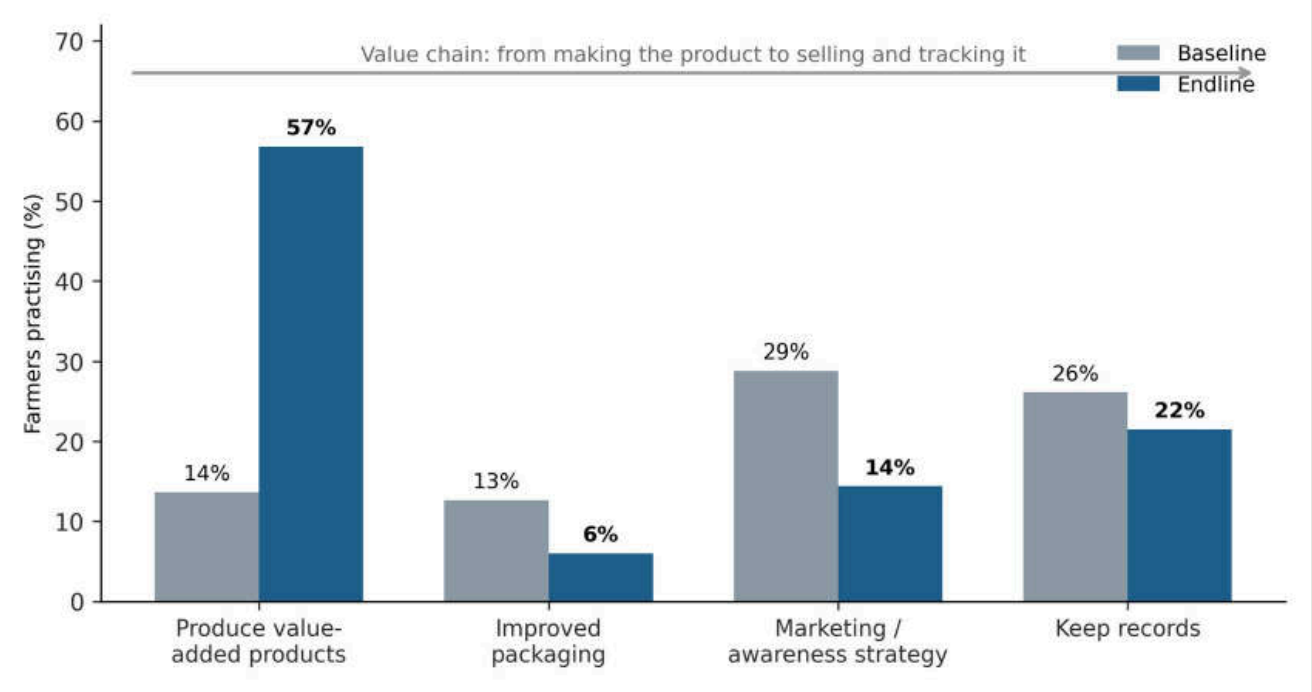


Figure 12. The value-added chain: production rose sharply, while packaging, marketing and record keeping fell.

The decline is mostly a denominator effect. Of the 75 farmers who now produce value-added products, only 8 (11%) package them, 19 (26%) market them and 24 (32%) keep records. Among farmers not doing each step, 39% to 45% have no value-added product at all, and another 8% to 21% make products only for the family (Figure 13).

Field notes add that those who tried packaging often made small amounts that were eaten immediately, groups packaged only during training, and packaging materials are costly. Farmers who do sell take the product to local markets themselves and see no need for a marketing strategy.

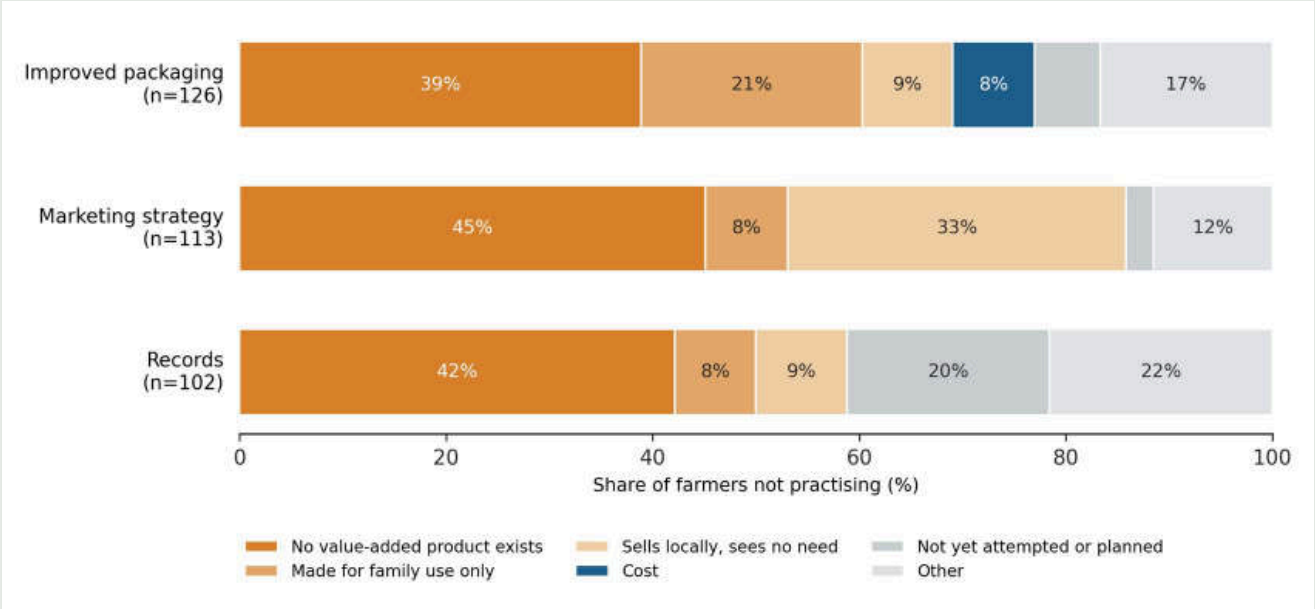


Figure 13. Reasons farmers were not practising packaging, marketing and records.

The barrier to production itself is not knowledge. Among the 57 farmers not producing value-added products, 23 plan to start or have not yet tried, and 16 lack raw materials or equipment such as dryers. Those who do produce often do so in groups that share equipment.

IMPLICATION

This is a sequencing problem, not a training gap. Farmers cannot package, market or record a product they do not yet make regularly or in volume.

ACTIONS

→ Organise group-level value addition where possible, so that farmers share equipment and produce in volume.

→ Keep emphasising general record keeping, even where value addition is not yet regular.

DAY 4

Livestock production

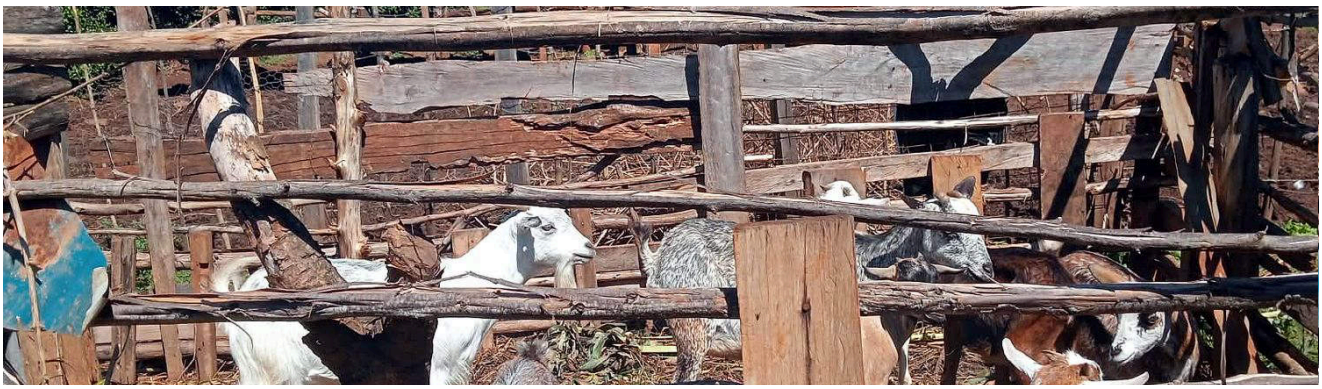


Photo 7. *Goat enterprise of an MHAC farmer in Kiminini, Trans Nzoia County, June 2026. Photo credit: Michael Chariga.*

Day 3 ended with farmers short of materials and equipment for value addition. Day 4 shows the same two lessons in livestock. Practices that rely on what the farmer already has, such as natural behaviour, exchanging males and botanical remedies, grew strongly. Practices that rely on bought or hard-to-find inputs, such as azolla and fermented feed, grew slowly or fell. Day 4 also adds a third lesson, about records. Figure 14 shows the picture.

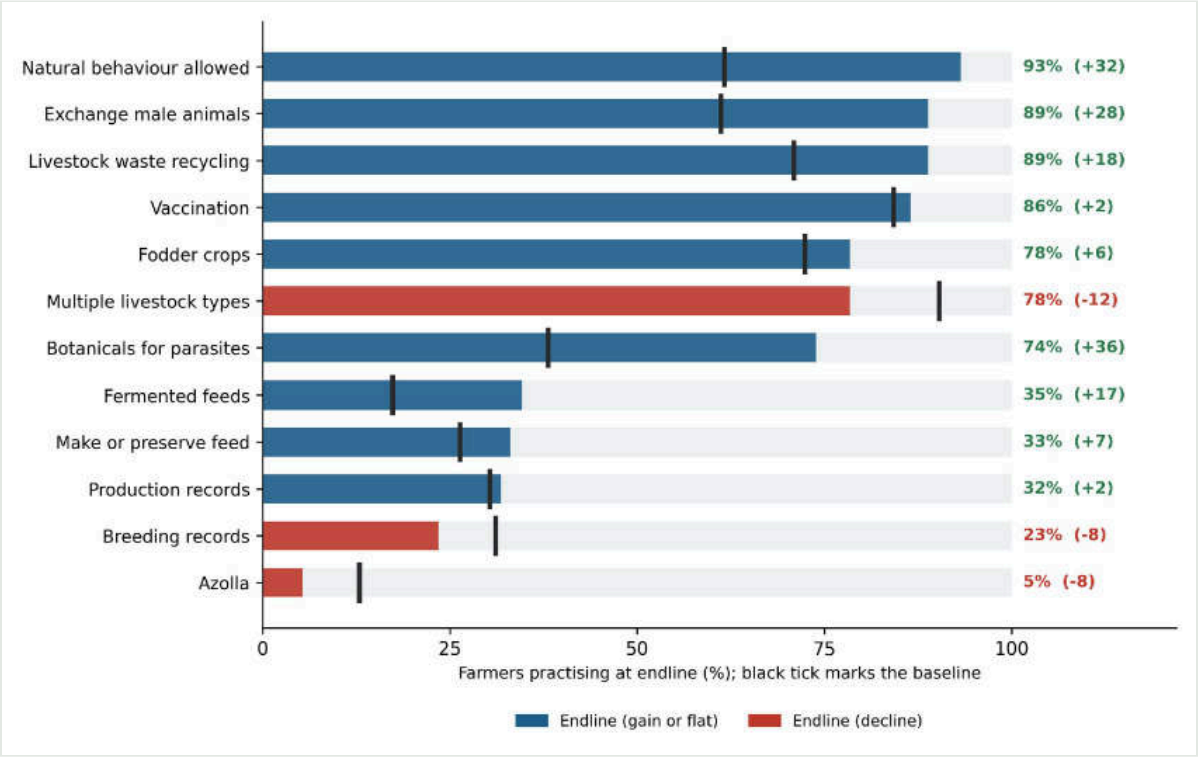


Figure 14. Share of farmers practising each Day 4 practice at endline; the black tick shows the baseline.

The great majority of farmers keep several types of livestock, and almost everyone grows fodder, with Napier[16] grass the most common. Keeping multiple livestock types is nonetheless one of the practices that fell, from 121 to 105 farmers (90% to 78%). Of the 29 farmers not keeping several types, 17 had too few animals or a narrow range, and 11 kept a single species, because of disease outbreaks, theft or selling the animals (Figure 15, left).

16 **Napier grass.** A tall, fast-growing fodder grass for livestock, also planted as hedgerows.

Finding 1: Practices that use local resources and knowledge grew strongly

Allowing natural behaviour rose from 82 to 124 farmers (62% to 93%), exchanging males to avoid inbreeding from 82 to 119 (61% to 89%), livestock waste recycling from 95 to 119 (71% to 89%) and botanicals for parasite control from 51 to 99 (38% to 74%, with 58 new farmers). Vaccination, already common, edged up from 84% to 86%.

Among the 35 farmers not using botanicals, 22 (63%) use purchased veterinary drugs instead (Figure 15, right). Fourteen of the 18 farmers not vaccinating prefer local treatment. Field notes add that most farmers say they have made herbal remedies such as aloe vera, tithonia, hot pepper and tephrosia, but few had any on hand during the visit, so the botanical figure may understate real use.

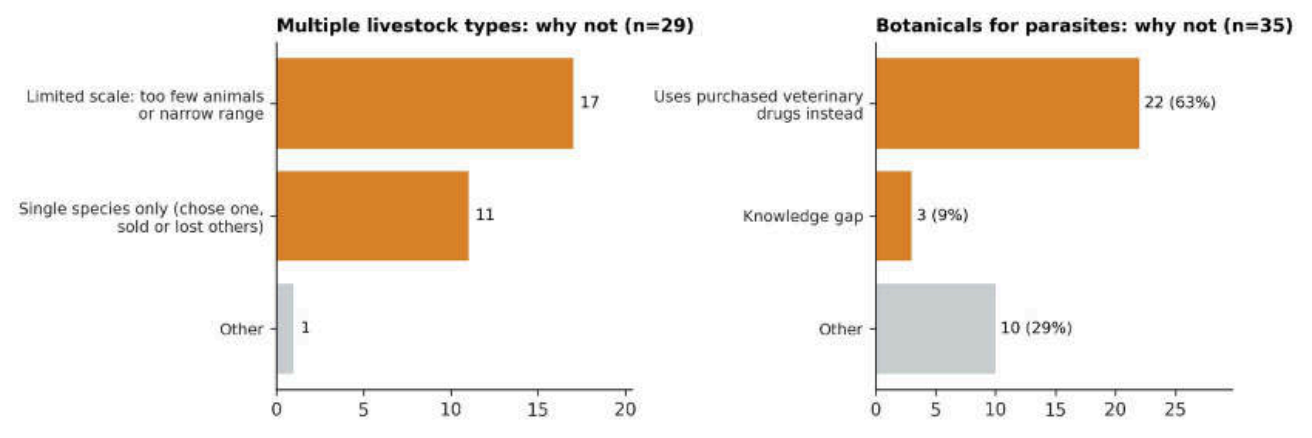


Figure 15. Left: reasons farmers were not keeping multiple livestock types. Right: reasons farmers were not using botanicals for parasites.

IMPLICATION

These practices are easy to adopt because they cost little. Purchased drugs, not lack of knowledge, are the main alternative to botanicals.

ACTIONS

→ Continue the current training on these practices.

→ Present botanicals as a complement to treatment, and keep the vaccination message clear, since many farmers rely on local treatment alone.

Finding 2: Feed preservation is held back by inputs, tools and scale

Making fermented feed rose from 23 to 46 farmers (17% to 35%), and making or preserving any feed rose from 35 to 44 (26% to 33%). Among the 87 farmers not making fermented feed, 27 (31%) cited access to inputs, 14 (16%) a knowledge gap and 12 (14%) few or no livestock (Figure 16). Preserving feed in general also ran into a habit: 14 farmers feed fresh fodder rather than preserve it, and some keep a few free-range backyard chickens and see no need.

Field notes add practical detail. Fermented feed is tedious to make, mainly because leafy ingredients must be chopped and farmers have no chopping machine. Leafy materials are seasonal and hard to find in the dry season, and grains are expensive. Molasses is hard to find. Some farmers lack enough fodder to preserve, and others lack storage for silage.

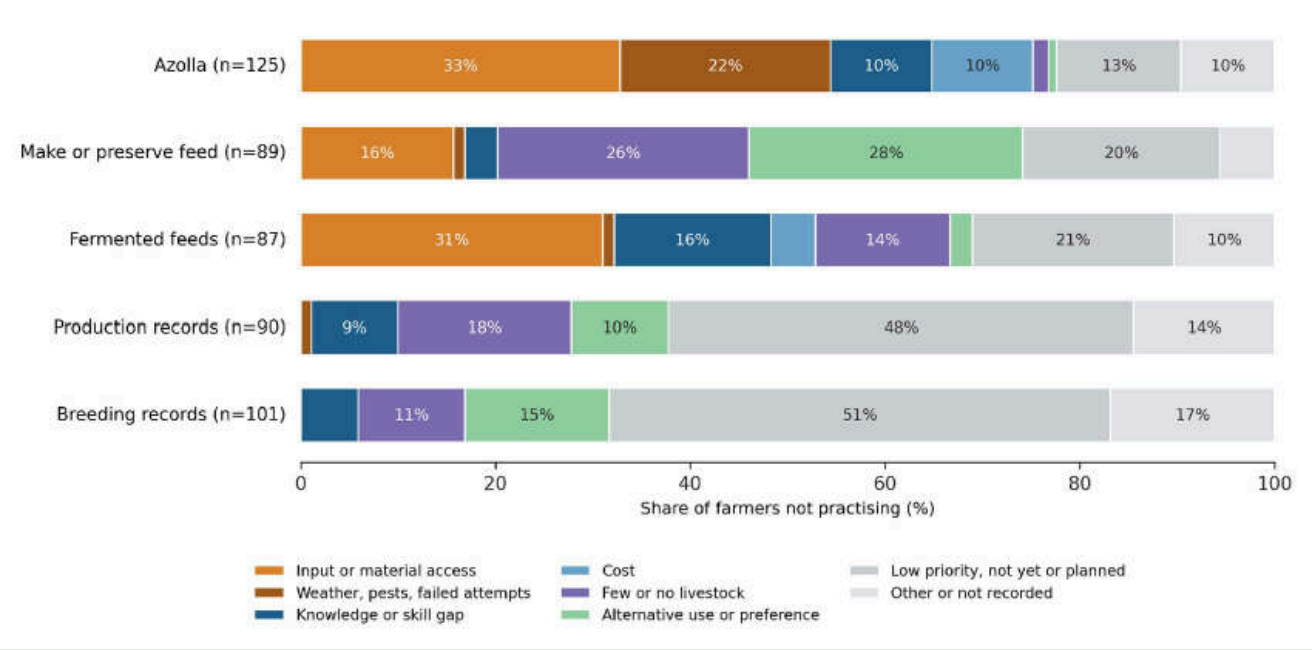


Figure 16. Reasons farmers were not practising the five weakest Day 4 practices, by share of those not practising.

IMPLICATION

Uptake is limited by inputs, labour and equipment, with a smaller role for knowledge. Molasses is also the bottleneck for bokashi on Day 1.

ACTIONS

→ Pool the work and equipment: introduce a shared chopping machine at group level.

→ Source molasses once for both bokashi and fermented feed, through farmer organisations.

Finding 3: Azolla is the weakest practice and needs a different approach

Azolla fell from 17 to 7 farmers (13% to 5%). Only 3 farmers started, 13 stopped and 112 never grew it. Among the 125 farmers not growing it, 41 (33%) cited access to materials, 27 (22%) said their attempt failed, 13 a knowledge gap and 13 cost (Figure 16).

Field notes explain the failures. Frogs invaded and destroyed the azolla in some cases. Others did not know how to manage shade, watering, hygiene or the dam liner, and their azolla dried. Farmers without azolla cited construction materials, not knowing how to build the pond, or not knowing where to find starter culture. Some also had security concerns about the pond.

IMPLICATION

The failures come from practical details and missing supply, and these are costly for a small farmer to learn alone.

ACTIONS

→ Rather than training every group, pilot azolla at a demonstration site and with a few interested groups, with a supply of starter culture.

→ Teach pond construction, shading, watering, hygiene and frog protection in a hands-on session.

Finding 4: Record keeping is not valued enough to be sustained

Keeping production records was flat, from 40 to 42 farmers (30% to 32%), and breeding records fell from 41 to 31 (31% to 23%), with 29 farmers stopping. Among the 101 farmers not keeping breeding records, 17 called it a low priority, 15 had not tried, 15 rely on memory, 14 see no need and 7 own no livestock. Production records follow the same pattern (Figure 16).

IMPLICATION

This is a perceived-value gap, not a skills gap. Farmers do not yet see what records give them. It matches the Day 3 finding on records for value-added products.

ACTIONS

→ Teach one simple page for each animal group, and tie records to something farmers already value, such as vaccination dates or the male exchange that is now at 89%, which depends on knowing a herd's breeding history.

DAY 5

Climate change adaptation



Photo 8. *A FIPS Africa farmer showing the terrace she built after the training, Butere, Kakamega County, June 2026. Photo credit: Michael Chariga.*

Day 5 closes the cycle by bringing trees, water and soil together, and two earlier threads return. The first is competition for crop residues from Day 1, which now shows up as a fall in mulching. The second is farmers' land, which decides whether practices such as terraces and contour planting are even needed. Eleven of 15 practices rose and four fell. Figure 17 shows how many farmers started and stopped each practice, which is more informative here than the net change, because so many practices saw farmers move in both directions.

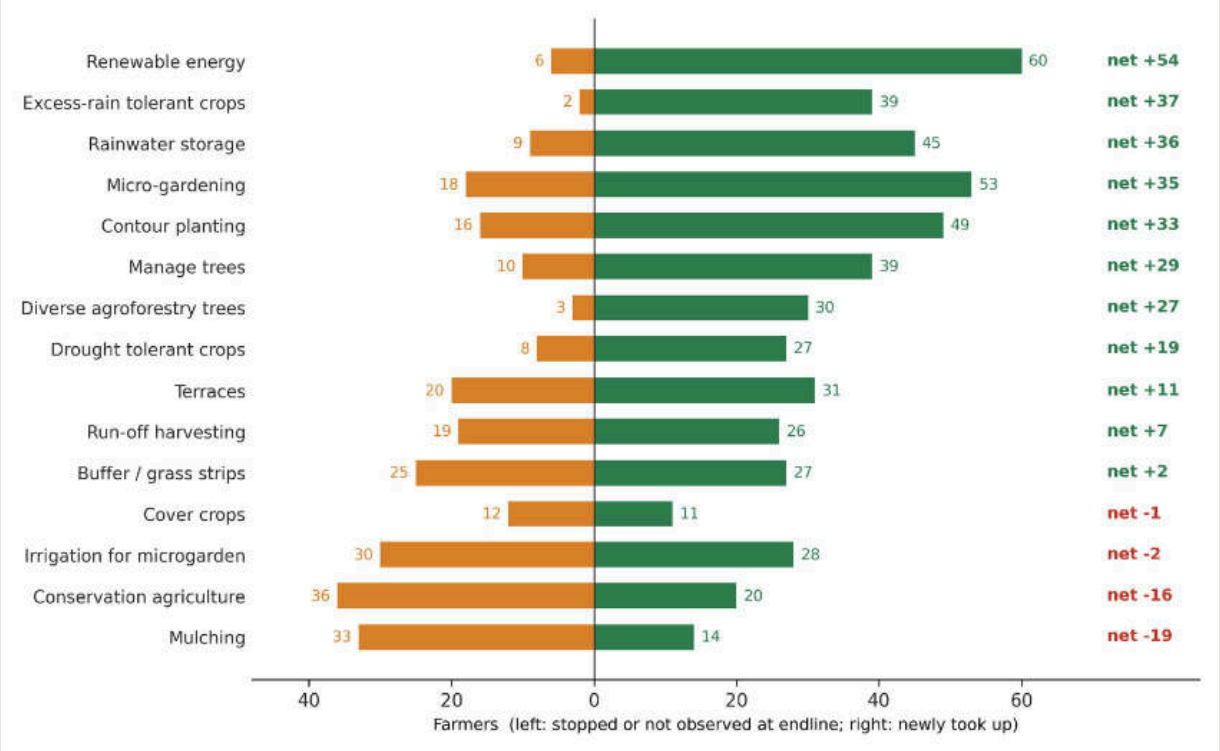


Figure 17. Farmers who stopped or were not observed at endline (left) and farmers who newly took up each Day 5 practice (right).

(right).

Finding 1: Trees, climate-resilient crops, energy and water storage grew strongly

Many Day 5 practices started high and went higher. Diverse agroforestry trees rose from 75% to 96% of farmers, managing trees from 65% to 87%(most of these trees however can not be attributed to training), crops tolerant of excess rain from 69% to 97% and drought tolerant crops from 79% to 93%. Green or renewable energy rose from 62 to 116 farmers (48% to 89%), with 60 new. Of the 14 farmers not using it, 13 rely on grid electricity. Rainwater storage rose from 66 to 102 farmers (50% to 77%), and of the 30 who do not store water, 20 use another water source such as a borehole, tap or river. Micro-gardening rose from 51 to 86 (39% to 66%), and contour planting from 54 to 87 (41% to 65%).

IMPLICATION

Where the practice fits farmers' needs and costs little, uptake is fast. The remaining non-practisers are mostly farmers with a good alternative, not farmers who lack knowledge.

ACTION

Future evaluations should have a way to track how training directly contributes to uptake of practices such as planting agroforestry trees.

Finding 2: Mulching and conservation agriculture fell because residues and tillage habits compete

Mulching fell from 107 to 88 farmers (81% to 67%), with 33 farmers stopping. Of the 44 farmers not mulching, 59% diverted crop residues to a competing use such as livestock feed or compost (Figure 18). This is the same story as Day 1 residues, where 19 of 33 farmers who stopped leaving residues sent them to livestock feed. Conservation agriculture[17] or reduced tillage fell from 77 to 61 farmers (59% to 47%), with 36 stopping. Among the 69 farmers not practising it, 35% cited a knowledge gap, 41% had not tried or planned to later, 8 prefer conventional ploughing and 3 fear more weeds.

Cover crops held almost steady (121 to 120 farmers), but the reasons for the 13 farmers not growing them echo Day 1: 7 grow only beans, which they harvest for food. Irrigating micro-gardens dipped slightly (55 to 53), with 30 stopping and 28 starting. Among the 79 farmers not irrigating, 36 have no micro-garden and 25 depend purely on rain, so many of them have nothing to irrigate.

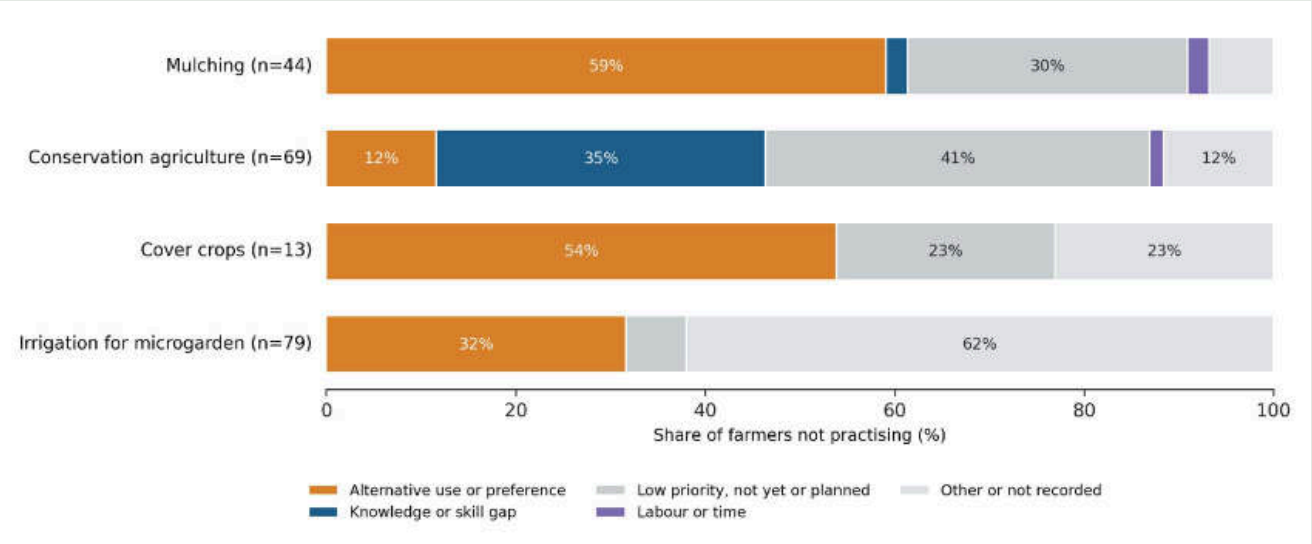


Figure 18. Reasons farmers were not mulching, practising conservation agriculture, growing cover crops or irrigating micro-gardens.

IMPLICATION

Mulching is falling because residues are in demand for livestock feed and compost, not because farmers dislike it. Conservation agriculture, by contrast, has a real knowledge gap. The irrigation figure is partly a measurement issue.

17 **Conservation agriculture.** Farming with minimum soil disturbance, permanent soil cover and crop rotation.

ACTIONS

→ Give one consistent message across Days 1 and 5: residues should return to the soil in some form, whether as mulch, compost or manure, and farmers can decide the route that fits their livestock.

→ Show conservation agriculture on a working plot, including how weeds are managed, to address the weed concern.

Finding 3: Run-off harvesting is a knowledge gap, while slope practices depend on the land

Run-off harvesting[18] (micro-catchments, zai pits[19] and sunken beds) rose only from 36 to 43 farmers (27% to 33%), and 69 never practised it. Of the 88 farmers not practising, 45% cited a knowledge gap and 39% had not tried or planned to later (Figure 19). This is the only water practice on Day 5 where training is the main barrier.

The slope-based practices tell a different story. Among farmers not using buffer or grass strips, contour planting or terraces, 62%, 89% and 80% respectively cited land or space: flat land with no erosion seen, small plots, or no control over land use. Field notes agree: because of land limits, buffer strips were uncommon, and the one farmer who had one planted a fodder crop without thinking of it as a buffer. Vegetative hedgerows, mainly Napier, were the most common measure that farmers carried out intentionally and could explain, followed by drainage channels that farmers dug on their own.

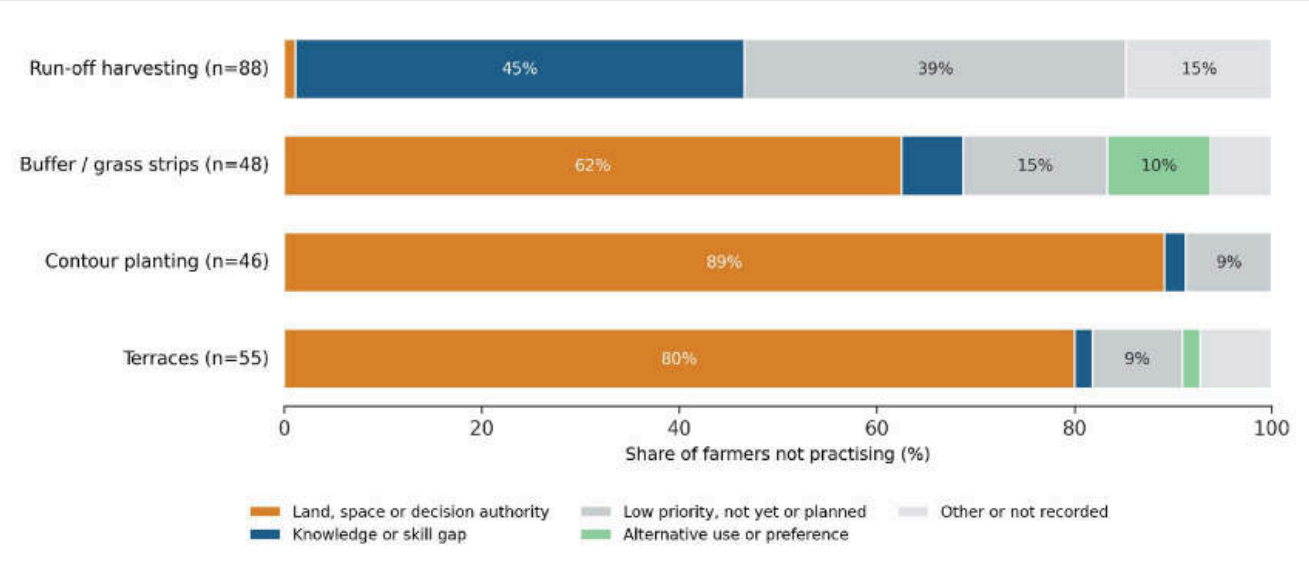


Figure 19. Reasons farmers were not practising run-off harvesting, buffer strips, contour planting or terraces.

IMPLICATION

For run-off harvesting, more training and practice would help. For slope practices, low uptake often reflects farms where the practice is not needed, so the result does not necessarily reflect a training failure.

ACTIONS

→ Run hands-on sessions for micro-catchments, zai pits and sunken beds.

18 **Run-off harvesting.** Capturing rainwater that flows over the land so it soaks in where crops grow.
19 **Zai pits.** Planting pits that collect rainwater and hold compost close to the crop.

Finding 4: Terrace uptake follows partner support more than training

Terraces rose from 65 to 76 farmers (50% to 58%), but the pattern follows who was supporting farmers. Field notes show that most farmers visited under FIPs Africa in Kakamega had terraces, and the organisation is running an experiment comparing continuous terraces, broken terraces and unterraced fields. In Siaya, GIZ works with TEMBEA Youth to support farmers financially to build terraces.

The data fit this. FIPs Africa farmers moved from 75% to 81% and TEMBEA farmers stayed at 75% (16 farmers each). Where baseline was already high, endline did not change, so the terraces predate the training. Siaya was 12 of 16 farmers at both rounds, not all of them. Gains were larger elsewhere, such as RFRN (25% to 62%, 8 farmers), CREPP (17% to 67%, 6 farmers) and MHAC (31% to 54%, 26 farmers). These group sizes are small, so they are indicative only (Figure 20).

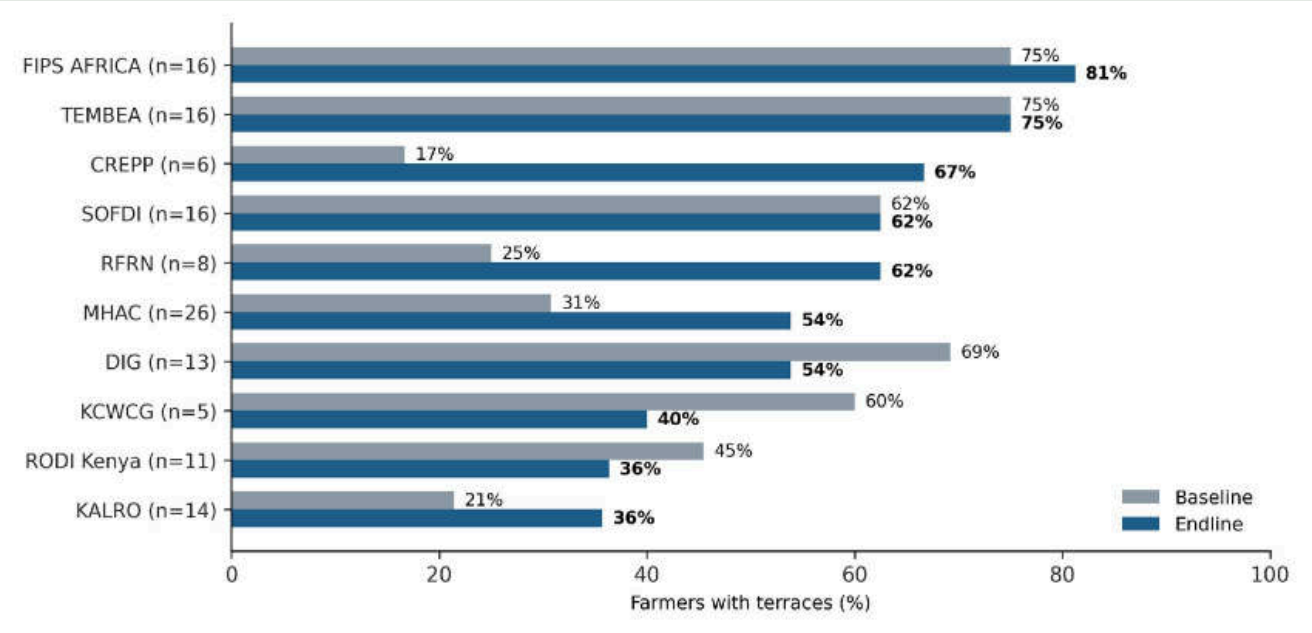


Figure 20. Share of farmers with terraces at baseline and endline, by partner organisation (organisations with five or more farmers).

IMPLICATION

Terraces are expensive to build, so adoption depends on organisations that provide direction and funding, not on training alone.

ACTIONS

→ Capture the FIPs Africa terrace experiment results and share them with other partners.

→ Where partners have funds, link terrace building to the training so that farmers leave with direction and support.

Across the programme: what Days 1 to 5 show together

Figure 21 shows the largest gains and losses across Days 2 to 5. The winners share a profile: they use local plants, knowledge or materials and have a visible benefit. The losers are mostly practices that compete with something else farmers need, such as a food crop, or that depend on conditions farmers do not control, such as weather and birds.

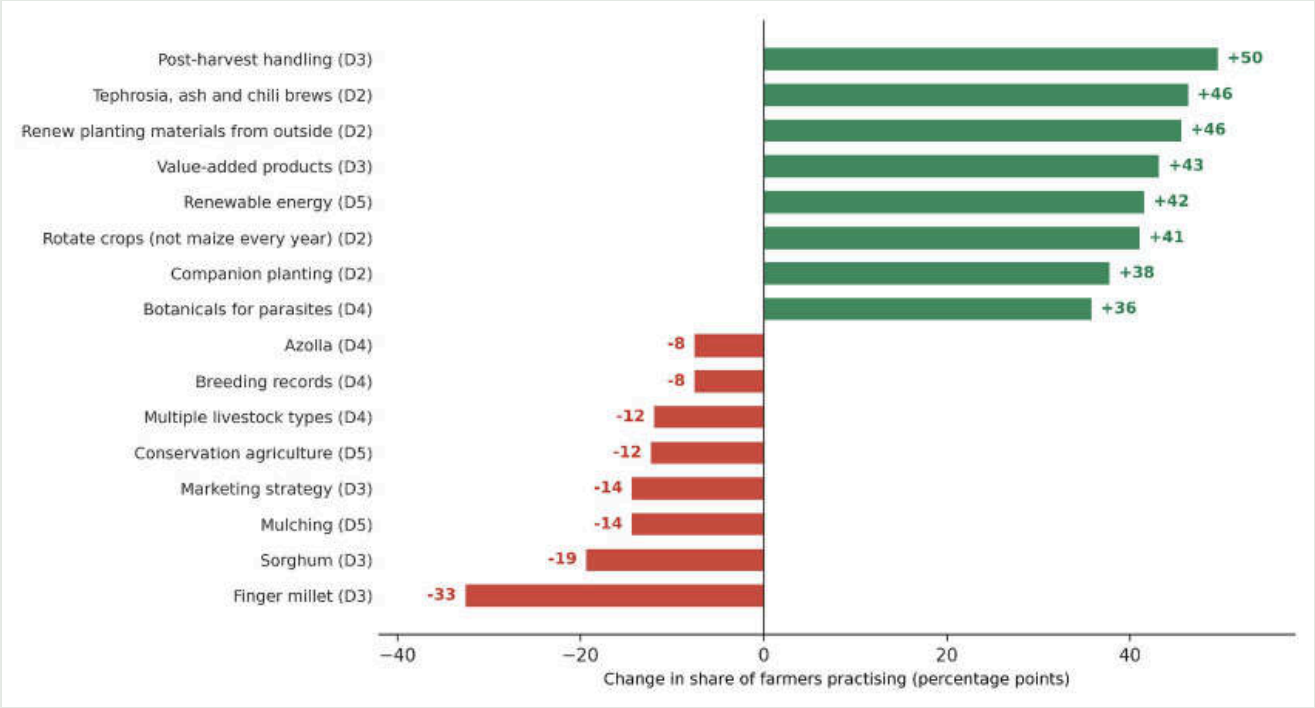


Figure 21. The eight largest gains and eight largest losses across Days 2 to 5, in percentage points.

Figure 22 compares what stopped farmers on each day. Reasons are grouped into common categories and shown as a share of that day's non-practising responses. Day 2's reverse-worded seed questions are excluded because "not doing" there means using own seed.

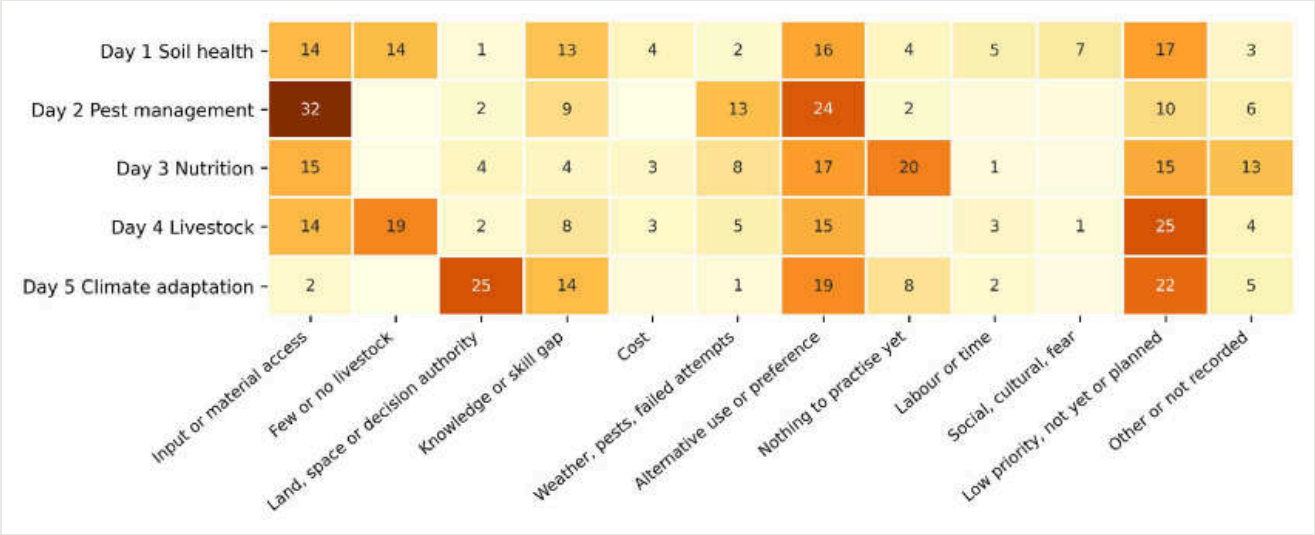


Figure 22 Reasons given for not practising at endline, by day, as a share of non-practising responses.

Farmers not found at endline

The evaluation used a census approach, so every farmer who attended the workshop was meant to be visited again,with exception of three farmers from Nandi county. In practice, not everyone could be found. Across the five days, 223 distinct farmers were surveyed at baseline and 141 (63%) were found at endline, leaving 82 farmers (37%) who were never assessed. Because some farmers who were reached did not have a complete record for every day, the matched samples used in the analysis range from 131 to 136 farmers per day, which is about 60 to 62% of the baseline sample. This section explains who was missing and why, because it shapes how far the results can be generalised.

Who was not found, and why

Evaluators recorded a reason for 53 of the 82 farmers who were not found (Figure 23). Two reasons stand out, and both point to young people. Fourteen farmers had left

home for work or were employed away from the farm: they had moved to Nairobi and other towns, taken jobs in gold mining or at a sugar company, or worked fixed office hours that left no time for an interview. Eight had returned to school or enrolled in a polytechnic, college or university. Together, these 22 farmers are 27% of all those not found, and 42% of those whose reason was documented. The evaluation team noted that many of these farmers were youth.

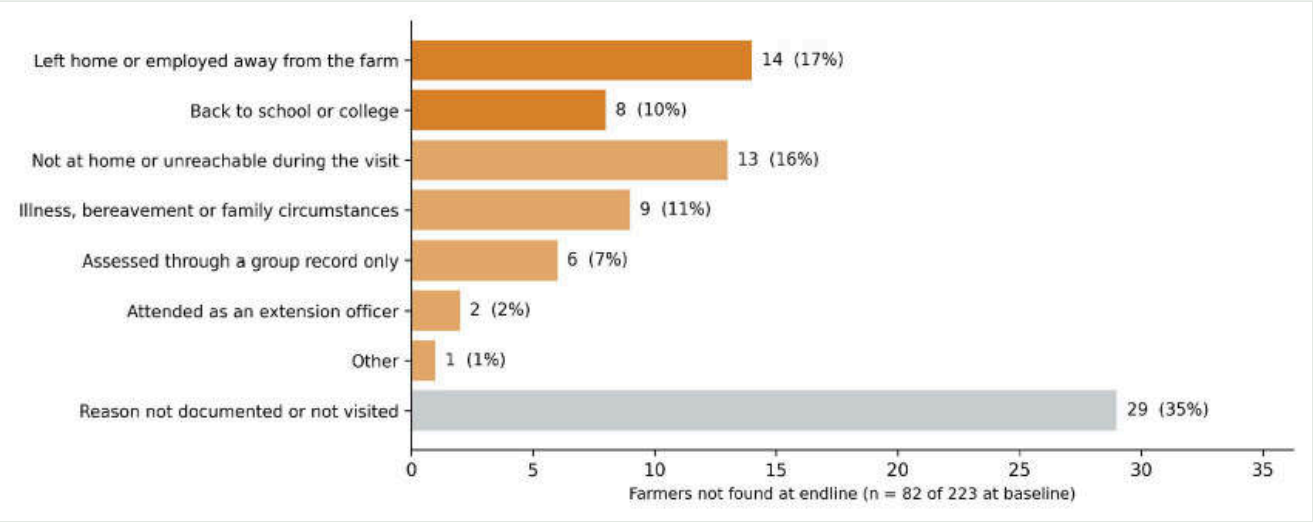


Figure 23. Reasons recorded for the 82 farmers (of 223 at baseline) who were not found at endline.

The other recorded reasons were mostly practical. Thirteen farmers were not at home or could not be reached during the visit: some leave early and return late, one had no phone, and others were at meetings or trainings. Nine were affected by illness, a bereavement, caring for a sick child or relative, or family conflict, including a separation from a spouse. Six farmers from one group were assessed through a single group record, and two attended the workshop as extension officers rather than as farmers. For 29 farmers (35%), the reason was not documented or the farmer was not visited at all.

Youth and women

The dataset does not record age or sex, so the number of youth and women among those not found cannot be counted. What the evidence does show is consistent with the evaluators' field observations. Youth are drawn away by jobs and further education, and those who stay face limited access to land, money and materials, which keeps their implementation low. Women and youth also often have no authority over land use and cannot persuade a spouse or parent to adopt a practice. For some women, the reasons recorded (caring for a sick child, a family separation) point to caregiving and household circumstances that made them harder to reach. Both groups therefore face constraints that affect whether they attend the follow-up, and whether they can apply what they learned.

Differences between partners and counties

Retention varied widely (Figure 24). TEMBEA and SOFDI found 80% of their baseline farmers (16 of 20), while KCWCG found 42% (5 of 12), RFRN 42% (8 of 19) and CREPP 37% (7 of 19). By county, 33% of Kisumu farmers (6 of 18) and 42% of Migori farmers (8 of 19) were found, and none of the four Kericho farmers. Part of the difference reflects how the follow-up was organised, not only how mobile the farmers were. For example, six CREPP farmers were assessed through one group record, and several KCWCG farmers have no documented reason.

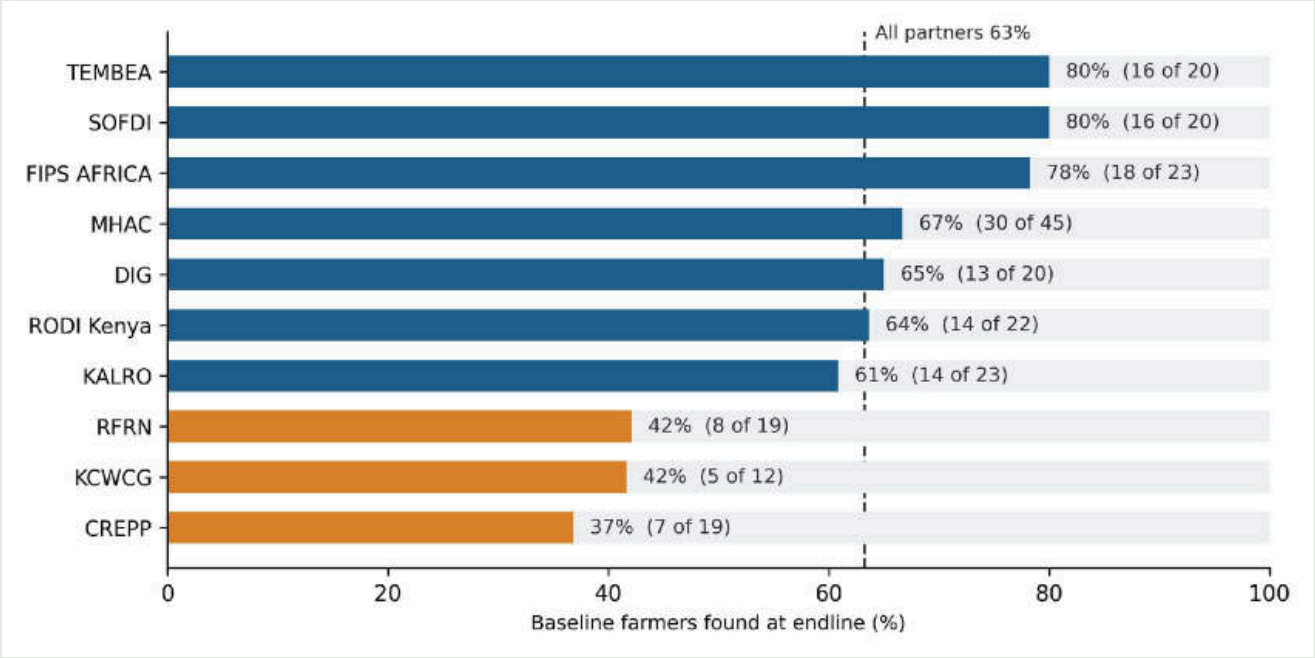


Figure 24. Share of baseline farmers found at endline, by partner organisation (223 farmers across the five days).

SECTION 3

Conclusions and Recommendations

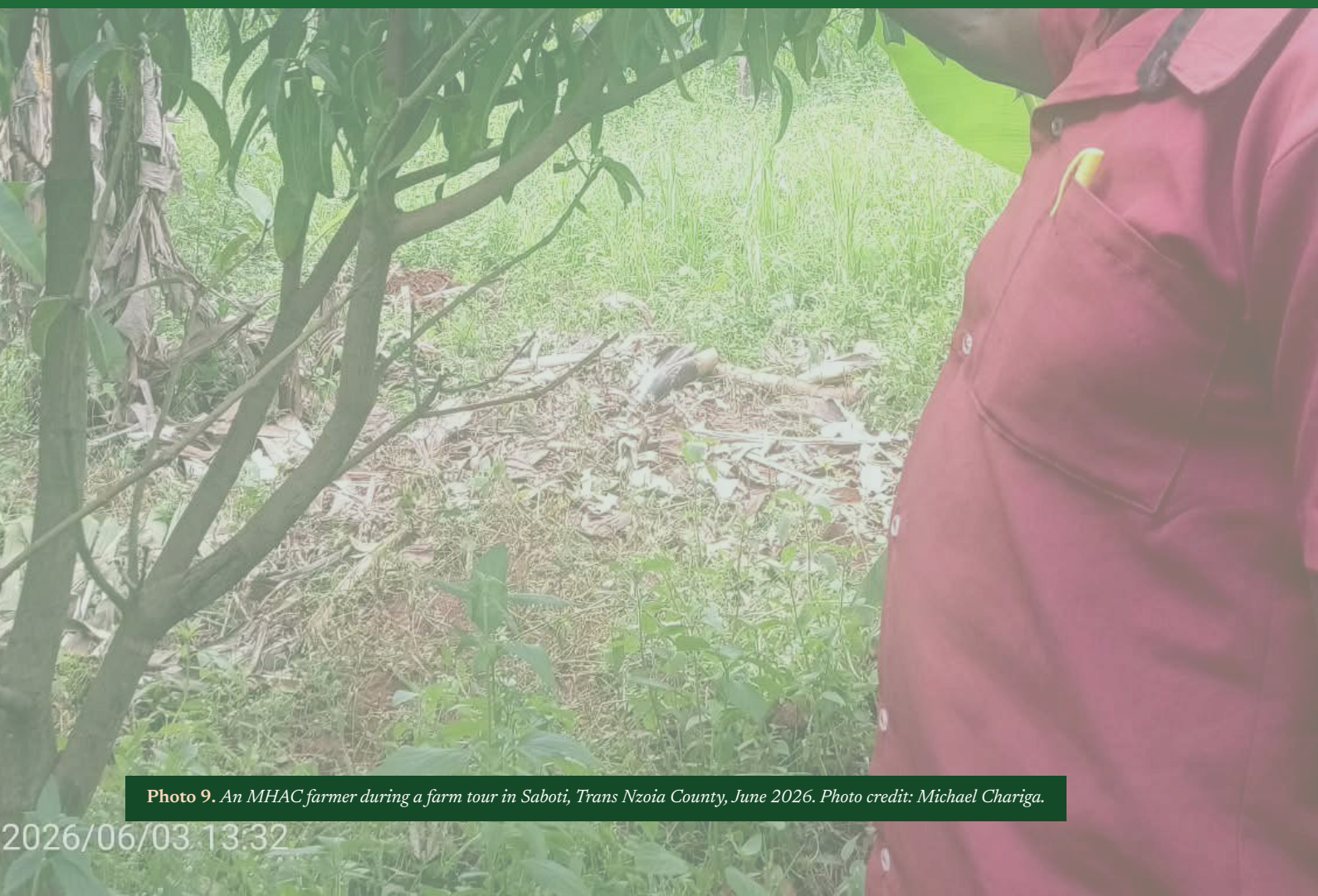


Photo 9. *An MHAC farmer during a farm tour in Saboti, Trans Nzoia County, June 2026. Photo credit: Michael Chariga.*

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Conclusions

The training produced real and widespread change on farms but not all changes can be attributed to the impact of training . Of the 63 practices assessed across the five training days, 48 were practised by more farmers at endline than at baseline and 15 by fewer, two of which were essentially flat. Practices like post-harvest handling went up by +50 percent but closer look at techniques farmers used revealed that farmers were using approaches that were not taught during the workshop. Also, the project was not implemented in isolation, there are instances where other organizations are supporting similar activities or where partners are promoting specific practice as part of their program. In other cases like planting agroforestry trees, these trees had been planted long before even training.

There are "low-hanging fruit" practices that farmers easily took up: Composting, botanical brews, companion planting, crop rotation, post-harvest handling and natural livestock behaviour all grew because they cost little, rely on local materials and show a visible benefit. Practices that depend on bought, distant or scarce inputs grew slowly or fell back.

Where practices stalled, the cause was mostly not a communication gap by trainers. The evaluation set out to test whether non-implementation reflects poor communication, or a mismatch between what MHAC assumed was relevant and what farmers find achievable. The evidence points mainly to the second. Five barriers recur across the days:

- **Access to seed and inputs:** flower seed, new crop seed, vermicompost worms, bokashi molasses, azolla starter culture and fermented feed ingredients.
- **Competition for residues, food and seed:** residues going to livestock feed and compost, cover crops seen as "wasting food", and pumpkin seed kept for planting rather than eaten.
- **Assets and land:** too few livestock to collect urine, small or flat plots for slope practices, and no authority over land use, especially for youth and women.
- **Social norms:** stigma and fear around human urine.
- **Environment:** erratic rainfall, drought and bird damage, which drove the decline in finger millet and sorghum. Genuine knowledge and skill gaps exist, but they are narrower. They are most visible in crop rotation (knowing which crops are in the same family), run-off harvesting, conservation agriculture, managing new crops from planting to harvest, and azolla management.

Part of the apparent decline is a measurement effect, and some gains cannot be credited to training. The termite-only residue question, the striga-only question, the combined intercropping and polyculture question, value-chain questions counted for farmers with no product, irrigation counted for farmers with no micro-garden, and seasonal cereals recorded as "not observed" all make results look worse than the practice on the ground. In the other direction, high baselines for trees and terraces show that partner programmes, not the workshop, explain much of that adoption. The tool shows whether a practice is present, not how extensively it is applied. Even so, 84% to 98% of endline "practising" responses were confirmed by observation, so the direction of change is reliable.

Recommendations and Lessons Learnt

Process

Recommendations on the evaluation process:

- **Strengthen the tool.** Add a simple magnitude scale (for example, improvising the tool to copy TAPE scale of 0-3) so that extent can be measured alongside presence. Separate practice from purpose in the residue and striga questions, and split the intercropping and polyculture question into two.
- **Time the visits for the period when farmers have something to show.** Visits that came after harvest or after weather damage left practices unseen. Plan the follow-up around cropping
- **Keep the joint assessment model and strengthen partners capacity to do continued farmer visits.** Most farmers were only visited by partners once or when visited the objective was not around the workshop training they had attended. If possible, partners should be facilitated to support frequent visits.

Context

Lessons learnt about the general context:

- **Weather is now a major risk to every practice.** Rainfall is erratic, and farmers can no longer plan around it. Practices that depend on timing, such as flowers, companion planting and crops sensitive to dry spells, are lost easily. Drought tolerant crops and water-conservation practices should be central to the message.
- **Land and decision-making power limit who can adopt.** Tenure, leases and household authority restrict choices, especially for youth and women.
- **Seed and input markets are thin.** New and alternative crops are mostly available only at MHAC. Seed viability is uneven across regions, and tephrosia seed has been mixed between purple-flowered and white-flowered types.
- **Farmers who have already attended other workshops and who host demonstrations implement the most.** First-time farmers had lower adoption.
- **Follow-up visits make a clear difference.** There is a wide gap between farmers visited often by extension officers and those never visited since the workshop.

Programme objectives and Results

Evaluation objective	What the evaluation found
Reveal which AE practices were being implemented	48 of 63 practices rose. Gains were largest where practices are low-cost and local, including composting, botanical brews, rotation, post-harvest handling and natural livestock behaviour.
Explore barriers for practices not implemented	Five recurring barriers: access to seed and inputs, competition for residues and food, assets and land, social norms and environmental risk. Knowledge gaps were real but narrower.
Test for a disconnect between MHAC assumptions and farm reality	A partial disconnect. Content was often relevant but not always achievable (for example vermicompost, azolla, new crops and value addition without equipment). Some messages also need refining, such as crop families for rotation, residue use and records.

Recommendations for the programme:

- **Fix seed and other material access.** Provide starter packets of seed (flowers and new crops) at the workshop, tell farmers before they come that they will be able to buy seed, and arrange an affordable supply of worms, molasses and azolla starter culture. Revisit seed storage so that seed does not lose viability, and guard against mixing of tephrosia types.
- **Teach fewer practices more deeply.** Reduce the content and slow the pace, particularly for older farmers. Rotate small groups through practicals so that every farmer handles the materials, and avoid crowding around a single trainer.
- **Teach alternatives by demonstration.** Build the cheaper option next to the standard one (for example, two bokashi piles), run hands-on sessions for run-off harvesting, conservation agriculture, pond construction and vermicompost, and hold a farm tour of new crops before cooking sessions.
- **Address the practices farmers are moving away from.** Discuss bird control and varieties for finger millet and sorghum, since their decline conflicts with the climate message.
- **Pilot, then scale.** Test azolla and human urine on demonstration sites and small trials before wider teaching.Also, involving farmers in FRN for these could be a viable option.
- **Include youth and women deliberately.** Invite household decision-makers, and design roles and activities that fit youth who have little land, such as bokashi sales and group enterprises.

Partners

Recommendations to partners:

- **Follow up systematically.** Visit farmers within the first season, again before the evaluation, and where possible with a checklist of what to follow-up on. Farmers who are visited and linked to other projects implement more and become lead farmers.
- **Use lead and demonstration farmers.** Choose farmers with high implementation to host demonstrations and group meetings, and pair first-time farmers with them.
- **Support farmer groups to act together.** The five farmers from each organisation are expected to form a group. Where possible, use these groups to share equipment (choppers, value-addition tools), buy inputs in bulk (molasses, seed), and run community sites such as vermicomposting.
- **Prepare farmers to buy seed.** Tell them before the workshop that seed will be on sale so that they can bring money.
- **Share what your own programmes are doing.** Record support that predates or runs alongside the training, such as terraces, tree planting and financial support, and share results from experiments such as the continuous and broken terrace trial.
- **Link farmers to inputs and markets.** Help farmers reach seed suppliers, worm and culture sources, and local markets for value-added products.

Annexe

Annex 1: Participating organisations, counties and farmers involved

Organisation	County	Farmers at baseline	Farmers found at endline
Development In Gardening (DIG)	Kisumu and Homabay	20	13
Farm Inputs Promotions (FIPS Africa)	Kakamega and Busia	23	18
Kenya Agriculture and Livestock Research Organization (KALRO)	Homa Bay	23	14
Kipsaina Crane and Wetland Conservation Group (KCWCG)	Trans Nzoia	12	5
Manor House Agricultural Centre (MHAC)	Trans Nzoia	45	30
Rural Farmers Research Network (RFRN)	Migori	19	8
Resource Oriented Development Initiative (RODI Kenya)	Kakamega and Uasin Gishu	22	14
Sustainable Organic Farming and Development Initiative (SOFDI)	Kakamega and Vihiga	20	16
TEMBEA	Siaya	20	16
Community Rehabilitation and Environmental Protection Program (CREPP)	Kisumu	19	7
Total	**10 counties, 42 sub-counties**	**223**	**141**

Annex 2: Partner organisations

Development In Gardening (DIG)

Farm Inputs Promotions (FIPS Africa)

Kenya Agriculture and Livestock Research Organization (KALRO)

Kipsaina Crane and Wetland Conservation Group (KCWCG)

Manor House Agricultural Centre (MHAC)

Rural Farmers Research Network (RFRN)

Resource Oriented Development Initiative (RODI Kenya)

Sustainable Organic Farming and Development Initiative (SOFDI)

TEMBEA

Community Rehabilitation and Environmental Protection Program (CREPP)

Annex 3: Supporting organisations

Manor House Agricultural Centre (MHAC)

INNODEMS

IDEMS

Annex 4: Funder

Conservation Food and Health Foundation (CFHF)

Annex 5: Notes on terminology

1. Agroecology (AE). Farming that applies ecological principles and works with local resources, biodiversity and farmers' own knowledge rather than relying mainly on bought inputs.

2. Baseline and endline. The baseline assessment recorded the practices farmers already used during the workshop. The endline assessment followed after farmers had about one and a half seasons to apply the practices.

3. Vermicompost. Compost made with the help of earthworms.

4. Azolla. A small floating water fern grown in ponds and used as livestock feed.

5. Census approach. Every targeted farmer is visited, instead of a sample of farmers.

6. TAPE. The Tool for Agroecology Performance Evaluation, which uses a scale to show how extensively a practice is applied.

7. Tephrosia. A legume shrub whose leaves and seed are used in botanical sprays against pests.

8. Matched farmers. Farmers with both a baseline and an endline record, so that their answers can be compared.

9. Cover crops and green manure. Crops grown mainly to protect and feed the soil, sometimes ploughed back before harvest.

10. Bokashi. Compost made by fermenting organic material with ingredients such as ash, biochar or molasses. It is ready faster than ordinary compost.

11. Tithonia. A bushy plant whose leaves are used as green manure or to make liquid fertiliser (tithonia tea).

12. Boma compost. Compost made from livestock manure and bedding collected in the animal enclosure (boma).

13. Striga. A parasitic weed that attacks cereal crops such as maize and sorghum.

14. GO, GROW and GLOW foods. Food groups used in nutrition education: GO foods give energy, GROW foods help the body grow and repair, GLOW foods protect health.

15. Hermetic storage bags. Airtight bags that keep out air and pests to protect stored grain and seed.

16. Napier grass. A tall, fast-growing fodder grass for livestock, also planted as hedgerows.

17. Conservation agriculture. Farming with minimum soil disturbance, permanent soil cover and crop rotation.

18. Run-off harvesting. Capturing rainwater that flows over the land so it soaks in where crops grow.

19. Zai pits. Planting pits that collect rainwater and hold compost close to the crop.